

5G RAN

Cell Management Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added compatibility of the pRRU5551JR with RF combination cells and supplementary cells. For details, see 3.3.3 Hardware .	None	High-frequency TDD	DBS5900 LampSite
None	None	Low-frequency TDD	None

- Editorial changes
 - Added descriptions of the cell frame offset. For details, see [3.1.3.8 Cell Frame Offset](#).
 - Revised the descriptions of differences between high and low frequency bands. For details, see [2.2 Differences](#).
 - Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Cell management	Supported in both NSA and SA networking, with the following differences: In SA or NSA and SA hybrid networking, the SSB frequency-domain position is configured by using global synchronization channel number (GSCN). In NSA networking, the SSB frequency-domain position is configured by using GSCN or NR-ARFCN. For details, see 3.1.3.5 SSB Frequency-Domain Position .	3 Cell Management

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Cell management	In typical scenarios, differences between high and low frequency bands are as follows: • Combined-RRU cells and split-sector cells are supported only in low frequency bands. • Supplementary cells are supported only in high frequency bands. • Cells in multi-carrier sectors and RF combination cells are supported in both high and low frequency bands. • For details about the differences between high and low frequency bands in other scenarios, see the related feature documents described in 3.1.2 Typical Scenarios. There are also differences in cell parameters, such as the cell frequency band, cell bandwidth, subcarrier spacing, and cell power.	3 Cell Management

3 Cell Management

3.1 Principles

Cell management refers to the management of communications resources by establishing cells and configuring related cell parameters.

3.1.1 Related Concepts

Cell-related concepts include the cells (NR cell and NR DU cell), sector, sector equipment, RF equipment, baseband equipment, NR DU cell transmission reception point (TRP), and NR DU cell coverage area. [Figure 3-1](#) illustrates an example mapping relationship between these concepts. In this example, one cell contains one NR DU cell TRP, and one NR DU cell TRP corresponds to one NR DU cell coverage area.

Figure 3-1 Cell-related concepts

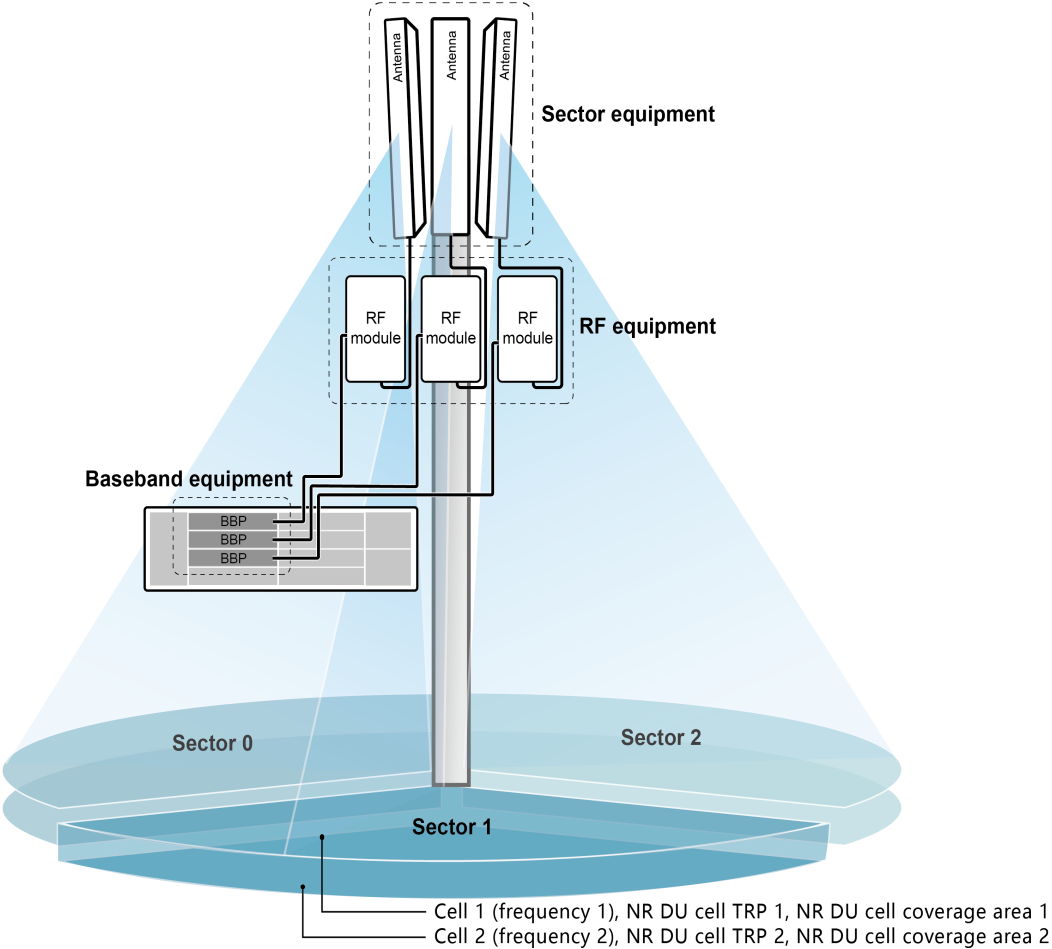
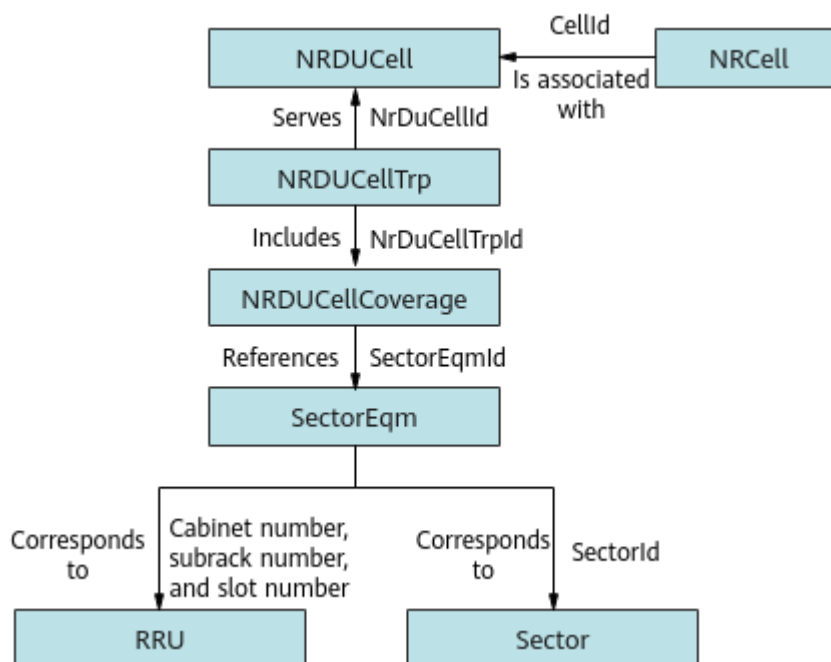


Figure 3-2 shows the relationships between concepts related to a common cell (with `NRDUCell.NrDuCellNetworkingMode` set to `NORMAL_CELL`).

Figure 3-2 Relationships between concepts related to a common cell



- One NR cell is associated with one NR DU cell, with **NRCell.CellId** and **NRDUCell.CellId** set to the same value.
- One NR DU cell TRP serves one NR DU cell, with **NRDUCell.NrDuCellId** and **NRDUCellTrp.NrDuCellId** set to the same value.
- One NR DU cell TRP includes one NR DU cell coverage area, with **NRDUCellTrp.NrDuCellTrpId** and **NRDUCellCoverage.NrDuCellTrpId** set to the same value.
- One NR DU cell coverage area references one piece of sector equipment, with **NRDUCellCoverage.SectorEqmId** and **SectorEqm.SectorEqmId** set to the same value.
- One piece of sector equipment is associated with one or two pieces of RF equipment, with the same cabinet number, subrack number, and slot number:
 - For the cabinet number: **SectorEqm.RRUCN** set to the value of **RRU.CN**
 - For the subrack number: **SectorEqm.RRUSRN** set to the value of **RRU.SRN**
 - For the slot number: **SectorEqm.RRUSN** set to the value of **RRU.SN**
- One piece of sector equipment is associated with one sector, with **SectorEqm.SectorId** and **Sector.SectorId** set to the same value.
- One piece of baseband equipment serves one or more NR DU cell TRPs, with **NRDUCellTrp.BasebandEqmId** and **BasebandEqm.BasebandEqmId** set to the same value.

3.1.1.1 Cell

A cell is an area where radio communications services are provided. It is the fundamental unit of a radio network. Coverage of the entire radio network is provided by cells.

According to section 3.1 "Definitions" in 3GPP TS 38.401 V15.4.0, a base station is logically divided into the following two types of nodes in the 5G New Radio (NR) system.

- A gNB central unit (gNB-CU) provides functions of the following layers:
 - Radio resource control (RRC) layer
 - Service Data Adaptation Protocol (SDAP) layer
 - Packet Data Convergence Protocol (PDCP) layer
- A gNB distributed unit (gNB-DU) provides functions of the following layers:
 - Physical layer
 - Radio link control (RLC) layer
 - Medium Access Control (MAC) layer

An NR cell and an NR DU cell must be configured separately, and then associated to form a gNodeB cell.

- NR cell: A traditional logical cell configured on the gNB-CU, for which basic cell information, such as the cell ID, cell name, frequency band, and duplex mode, are configured.
- NR DU cell: A physical cell configured on the gNB-DU, to which baseband and RF resources are allocated.

Cell-Related Operations

[Table 3-1](#) lists the cell-related operations.

Table 3-1 Cell-related operations

Operation Type	MML Command	Description
Add NR Cell	ADD NRCELL	This command is used to add an NR cell.
Add NR DU Cell	ADD NRDUCELL	This command is used to add an NR DU cell. One NRCell MO corresponds to one or two NRDUCell MOs. For example, for a common cell, one NRCell MO corresponds to one NRDUCell MO; for a supplementary uplink (SUL) cell, one NRCell MO corresponds to two NRDUCell MOs. For details about common cells and SUL cells, see 3.1.2.1 Common Cell and 3.1.2.3 SUL Cell , respectively.
Activate NR Cell	ACT NRCELL	This command is used to activate an NR cell.

Operation Type	MML Command	Description
Deactivate NR Cell	DEA NRCELL	This command is used to deactivate an NR cell. A cell must be deactivated, for example, in scenarios where the cell needs to be reconstructed. This operation causes service drops in the cell, affecting user experience. To minimize the probability of service drops, the corresponding NR DU cell can be blocked before NR cell deactivation.
Block NR DU Cell	BLK NRDUCELL	This command is used to block an NR DU cell. After a cell is blocked, new UEs are not allowed to access or be handed over to this cell. The NRDUCell.CellAdminState parameter specifies the priorities of NR DU cell blocking as follows: • CELL_LOW_BLOCK (low priority): If there are no RRC_CONNECTED UEs in the cell, the cell is deactivated immediately upon command execution. If there are RRC_CONNECTED UEs in the cell, the base station checks the number of these UEs every 5s and deactivates the cell once the number is 0. • CELL_MID_BLOCK (medium priority): The mechanism is similar to that of the low priority, except that served UEs need to be transferred out of the cell within a duration of 60s after the MML command is executed. The cell is forcibly deactivated after 60s even if RRC_CONNECTED UEs still exist in the cell. • CELL_HIGH_BLOCK (high priority): The mechanism is similar to that of the medium priority, except that the preceding duration is shortened from 60s to 10s.

 NOTE

NSA UEs are transferred out of a cell through the RRC connection release procedure, while SA UEs are transferred out of a cell through the RRC redirection procedure.

In high-frequency TDD, operations such as activating or deactivating an NR cell and blocking an NR DU cell will change the number of cells in a sector. In addition, possible cell coverage changes will lead to fluctuations in the values of KPIs such as those related to the number of UEs in a cell, UE throughput, access, handovers, and service drops.

Cell-Level Networking Option

The networking option of a cell is determined by the networking option of the cell operator (specified by the **gNBOperator.NrNetworkingOption** parameter) and the cell-level networking option (specified by the **NRDUCellOp.NrNetworkingOption** parameter) as follows:

The value of the **gNBOperator.NrNetworkingOption** parameter is considered in the first place.

- If the value is **SA**, the networking option of the cell is SA only.
- If the value is **NSA**, the networking option of the cell is NSA only.
- If the value is **SA_NSA**, the value of the **NRDUCellOp.NrNetworkingOption** parameter is further considered.
 - If the value is **SA**, the networking option of the cell is SA only.
 - If the value is **NSA**, the networking option of the cell is NSA only.
 - If the value is **SA_NSA** or **INVALID**, the networking option of the cell is NSA and SA hybrid networking.

NOTE

If the **NRDUCellOp.NrNetworkingOption** parameter is set to **INVALID**, the networking option of the cell is the same as its operator, that is, indicated by the **gNBOperator.NrNetworkingOption** parameter.

Impact of Transmission Interface Faults on Cells

- **In SA-only networking**, an NR cell is automatically deactivated when the NG interface of the cell is faulty.
- **In NSA-only networking**, an NR cell is automatically deactivated in either of the following scenarios:
 - Scenario 1: The **S1_U_OM_FAULT_AUTO_DEA_SW** option of the **gNodeBParam.CellStatePolicySwitch** parameter is selected, the OM channel is faulty, and any of the following conditions is met:
 - The S1-U interface configured for the NR cell is faulty.
 - The S1-U interface configured for the NR cell is removed due to a fault.
 - No S1-U interface is configured and all X2-C interfaces are faulty for the NR cell.After the NR cell is automatically deactivated, it will be reactivated when none of the preceding conditions is met or when the base station restarts. In the latter case, the automatic deactivation conditions will be checked again once the OM channel is steadily connected.
 - Scenario 2: The NR cell is served by a co-MPT base station, the **COMPT_S1_U_UNAVAIL_AUTO_DEA_SW** option of the **gNodeBParam.CellStatePolicySwitch** parameter is selected for this cell, the S1-C interface has been configured for the LTE cell also served by this base station, and all the following conditions are met:
 - No S1-U interface is available for the NR cell. (No S1-U interface is configured or the S1-U interface is faulty.)
 - No X2-C interface is available for the NR cell. (No X2-C interface is configured or the X2-C interfaces are faulty.)
 - The S1-C interface of the LTE cell is faulty.

In local deployment scenarios, it is recommended that the **COMPT_S1_U_UNAVAIL_AUTO_DEA_SW** option of the **gNodeBParam.CellStatePolicySwitch** parameter be deselected.

 NOTE

- Static GTP-U echo must be enabled before the **S1_U_OM_FAULT_AUTO_DEA_SW** or **COMPT_S1_U_UNAVAIL_AUTO_DEA_SW** option of the **gNodeBParam.CellStatePolicySwitch** parameter is selected. For details about how to configure static GTP-U echo, see *IPv4 Transmission* or *IPv6 Transmission*.
- For an NR cell served by a co-MPT base station with the S1-C interface configured for an LTE cell also served by this base station, if the **S1_U_OM_FAULT_AUTO_DEA_SW** and **COMPT_S1_U_UNAVAIL_AUTO_DEA_SW** options of the **gNodeBParam.CellStatePolicySwitch** parameter are both selected, the NR cell is automatically deactivated when either of the preceding scenarios occur and reactivated when neither of the scenarios occur.
- **In NSA and SA hybrid networking**, an NR cell is automatically deactivated when any of the following conditions is met:
 - The NR cell deactivation conditions in SA-only networking and in NSA-only networking are all met.
 - The **NG_FAULT_OM_NORM_AUTO_DEA_SW** option of the **gNodeBParam.CellStatePolicySwitch** parameter is selected, the NG interface is faulty, and the OMCH is normal.
 - The **NG_OM_FAULT_AUTO_DEA_SW** option of the **gNodeBParam.CellStatePolicySwitch** parameter is selected, and both the NG interface and OMCH are faulty.

 NOTE

In the preceding scenarios, the cell can be automatically deactivated only after the anti-jitter time for interface fault detection.

In NSA-only networking as well as NSA and SA hybrid networking, automatic cell deactivation reduces the numbers of both the attempts and failures of SgNB changes from a neighboring base station to the local base station, which increases the SgNB change success rate on the entire network.

3.1.1.2 Sector

A sector is an area covered by an antenna. It uses one or more radio carriers to provide coverage and can be added by running the **ADD SECTOR** command. As described in [Table 3-2](#), a sector can be omnidirectional or directional, depending on the antenna type.

Table 3-2 Omnidirectional and directional sectors

Sector Type	Definition	Characteristic
Omnidirectional	An omnidirectional sector is a circular area covered by an omnidirectional antenna at its center.	In light-traffic areas, deploying omnidirectional sectors requires fewer antennas, which reduces equipment cost.

Sector Type	Definition	Characteristic
Directional	A directional sector is a sector area covered by a directional antenna. Multiple directional sectors can be deployed to cover a circular area. For example, if three or six sectors are configured for a gNodeB, each directional antenna covers a 120° or 60° sector, respectively. The actual azimuth of each antenna is slightly greater than the theoretical value so that the sectors overlap each other and provide seamless coverage.	• In heavy-traffic areas, deploying directional sectors supports more traffic volume. • Directional sectors support more flexible area and sector planning.

3.1.1.3 Sector Equipment

Sector equipment is a set of antennas used for the same sector. It is associated with a sector by configuring the **SECTOREQM** MO.

3.1.1.4 RF Equipment

RF equipment refers to remote radio units (RRUs) or active antenna units (AAUs). It is added and associated with CPRI ports on baseband processing units (BBPs) by running the **ADD RRUCHAIN** and **ADD RRU** commands, and is associated with sector equipment by running the **ADD RRU** and **ADD SECTOREQM** commands. For details about RF equipment, see *RF Unit and Topology Management*.

3.1.1.5 Baseband Equipment

Baseband equipment is a set of baseband signal processing units, which is used to process cell baseband data. A BBP can be added by running the **ADD BRD** command, and then be bound to a piece of baseband equipment by running the **ADD BASEBANDEQM** command.

3.1.1.6 NR DU Cell TRP

A TRP is an antenna array with one or more antenna elements available to the network. It is located at a specific geographical location for a specific area. For details, see section 3.1 "Definitions" in 3GPP TR 38.913 V14.2.0.

The basic information about an NR DU cell TRP comprises information about the baseband and RF equipment of the cell, which is used to specify the baseband equipment for the cell. If no baseband equipment is specified, the BBP directly connected to the RRU or AAU is preferentially used as the baseband equipment. An NR DU cell TRP is added by running the **ADD NRDUCELLTRP** command. For a common cell, one **NRDUCell** MO corresponds to only one **NRDUCellTrp** MO.

3.1.1.7 NR DU Cell Coverage Area

An NR DU cell coverage area corresponds to an NR DU cell TRP. It is associated with sector equipment by running the **ADD NRDUCELLCOVERAGE** and **ADD SECTOREQM** commands.

3.1.2 Typical Scenarios

Typical cell scenarios include common cells and cells in a multi-carrier sector.

3.1.2.1 Common Cell

A common cell is established in a sector served by an RF module, as shown in [Figure 3-3](#).

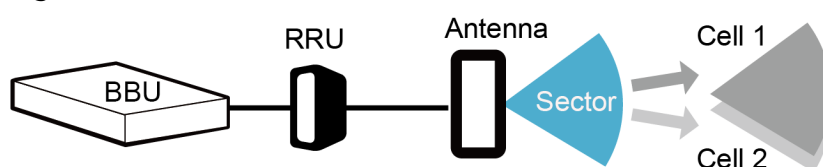
Figure 3-3 Common cell



3.1.2.2 Cells in a Multi-Carrier Sector

In this scenario, cells working on different frequencies are set up in the same sector, as shown in [Figure 3-4](#). This solution helps expand system capacity without deploying additional RF modules.

Figure 3-4 Cells in a multi-carrier sector



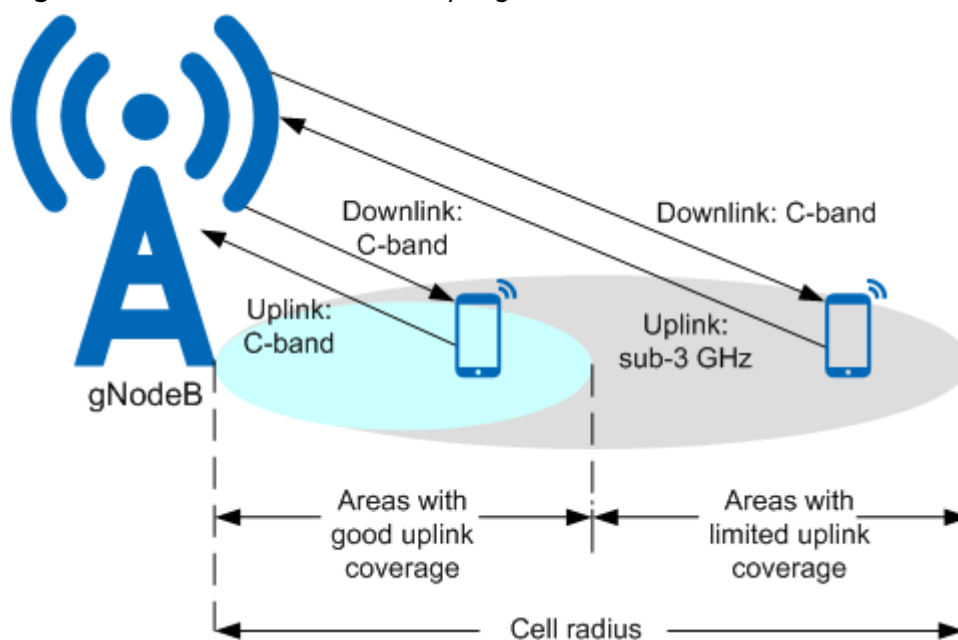
Cells in a multi-carrier sector apply to densely populated urban areas where traffic increases rapidly, such as exhibition centers, airport waiting halls, shopping malls, central business districts, and upmarket residential areas.

For details about the number of carriers supported by an RF module, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

3.1.2.3 SUL Cell

An SUL cell works in a sub-3 GHz band and is used only for uplink data transmission in UL and DL Decoupling scenarios. In such scenarios, C-band (such as 3.5 GHz) is used for the uplink and downlink, and a sub-3 GHz band (such as 1.8 GHz or 700 MHz) is used for the uplink to improve uplink coverage, as shown in [Figure 3-5](#). For details about UL and DL Decoupling and SUL cells, see *UL and DL Decoupling*.

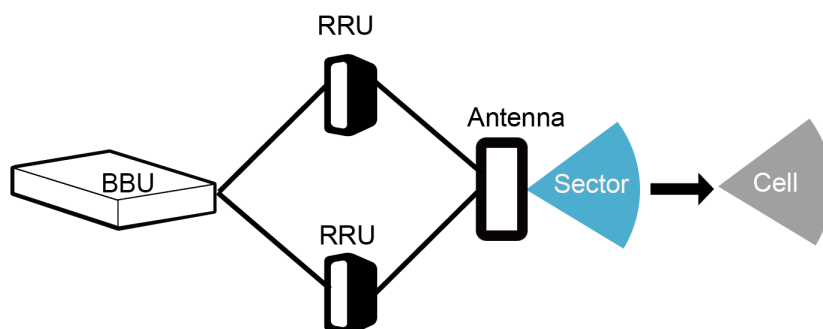
Figure 3-5 How UL and DL Decoupling works



3.1.2.4 Combined-RRU Cell

A combined-RRU cell is served by two RRUs connected to the same antenna. These RRUs correspond to the same sector and cover the same area. In this way, the cell capacity is improved. Combined-RRU cells are supported only by macro base stations, not by LampSite base stations.

Figure 3-6 Combined-RRU cell

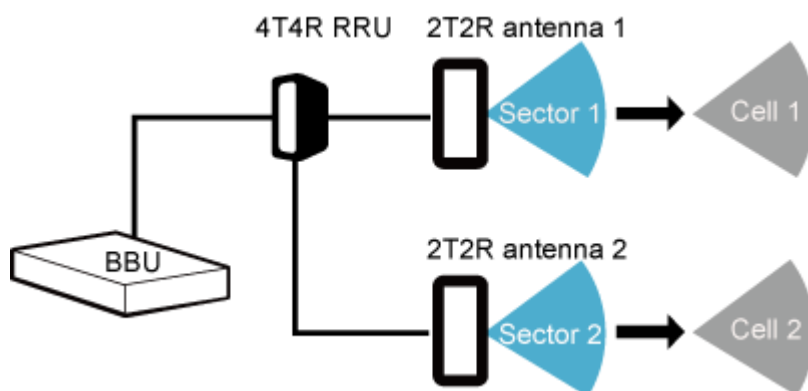


3.1.2.5 Split-Sector Cells

In RRU splitting scenarios, an RRU is connected to two antennas to serve two independent sectors. Each sector will use half of the antenna channels for transmitting and receiving signals. The cell corresponding to such a sector is called a split-sector cell.

Figure 3-7 shows how two split-sector 2T2R cells are set up with two antennas connected to a 4T4R RRU.

Figure 3-7 Setting up two split-sector 2T2R cells



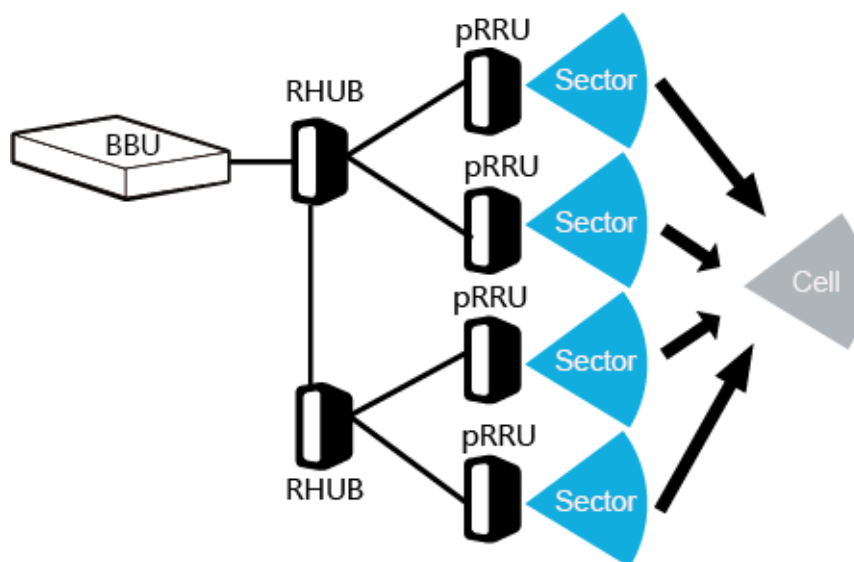
NOTE

In TDD, split-sector cells must have the same uplink and downlink frequencies, uplink and downlink bandwidths, slot configuration, frame offset, CPRI compression configuration, and uplink and downlink cyclic prefix (CP) lengths.

3.1.2.6 RF Combination Cell

RF combination cells are supported by LampSite base stations. RF signals from multiple pRRUs serving such a cell are combined by RHUBs, as shown in [Figure 3-8](#), extending cell coverage.

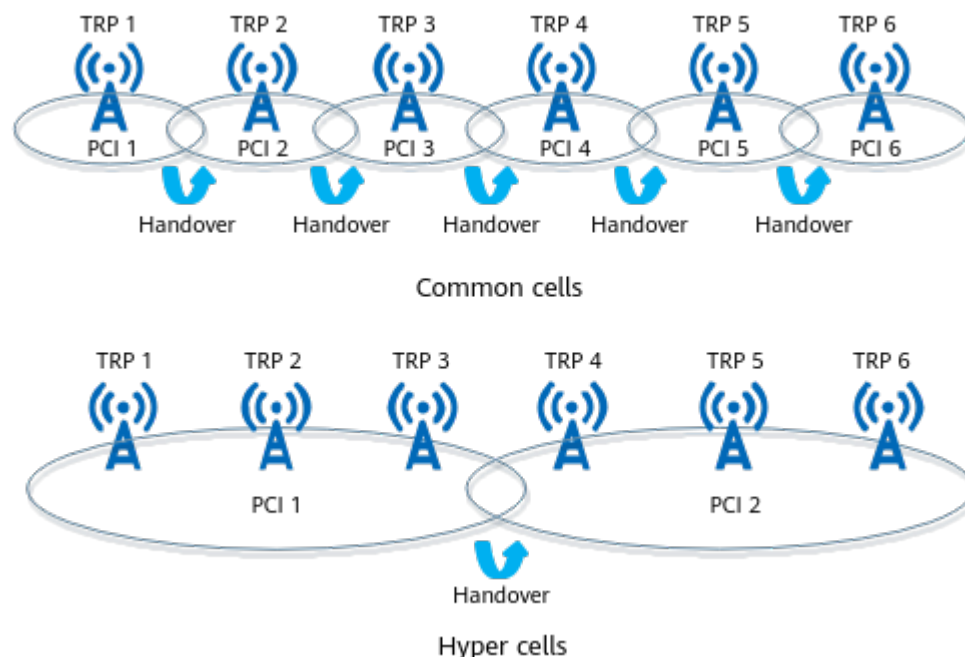
Figure 3-8 RF combination cell



3.1.2.7 Hyper Cell

A hyper cell is formed by combining multiple independent NR DU cell TRPs working on the same frequency and having the same bandwidth. It works as a logical cell to provide radio communication services for UEs, as shown in [Figure 3-9](#). For a hyper cell, one **NRDUCell** MO can be configured with multiple **NRDUCellTrp** MOs. For details about hyper cells, see *Hyper Cell*.

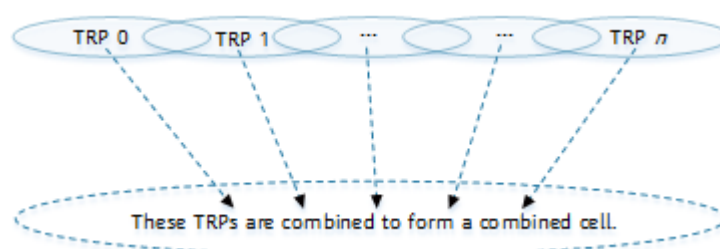
Figure 3-9 Example of common cells and hyper cells



3.1.2.8 Combined Cell

A combined cell is formed by combining multiple independent NR DU cell TRPs working on the same frequency and having the same bandwidth. It provides radio communication services for UEs as a whole, as shown in [Figure 3-10](#). For a combined cell, one **NRDUCell** MO can be configured with multiple **NRDUCellTrp** MOs. For details about combined cells, see *Cell Combination*.

Figure 3-10 Example of a combined cell



3.1.2.9 Distributed Massive MIMO Cell

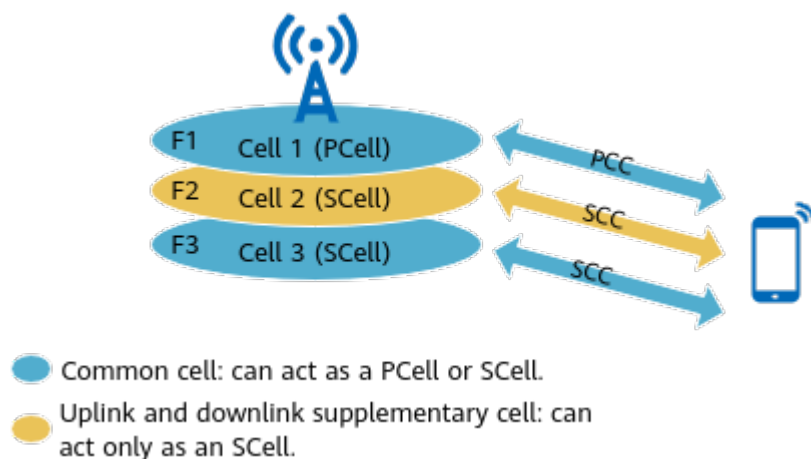
A distributed massive MIMO cell is a $4nT4nR$ massive MIMO cell formed by combining n $4T4R$ NR DU cell TRPs working in the same frequency band. For a distributed massive MIMO cell, one **NRDUCell** MO must be configured with multiple **NRDUCellTrp** MOs. For details about distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

3.1.2.10 Supplementary Cell

An uplink and downlink supplementary cell (referred to as a supplementary cell) has uplink and downlink channels. Such a cell can only serve as an SCell of CA to

improve uplink and downlink performance. A cell is an uplink and downlink supplementary cell if the **NRDUCell.SupplementaryCellInd** parameter is set to **ULDL_SUPPLEMENTARY_CELL** for this cell. This setting applies only in high frequency bands. **Figure 3-11** shows the uplink and downlink supplementary cell.

Figure 3-11 Uplink and downlink supplementary cell



As uplink and downlink supplementary cells do not support SSB transmission, they cannot be measured by UEs, and therefore UEs cannot camp on or access these cells.

During mobility-related parameter configuration, any uplink and downlink supplementary cell cannot be used as the target cell for blind handover or blind redirection. As such, it is recommended that no NR cell frequency relationship be added for the frequency of an uplink and downlink supplementary cell, or that the **NRCellFreqRelation.ConnFreqPriority** parameter be set to 0 for the frequency of an uplink and downlink supplementary cell.

3.1.3 Cell-related Parameters

This section describes key parameters related to the cell configuration, including the frequency band, NR Absolute Radio Frequency Channel Number (NR-ARFCN), and synchronization signal and PBCH block (SSB) frequency-domain position.

For details about other parameters of a cell, see the corresponding feature document. For example, for details about parameters related to CPRI compression, see *CPRI Compression*.

3.1.3.1 Frequency Band

The frequency band of a cell is specified by the **NRCell.FrequencyBand** and **NRDUCell.FrequencyBand** parameters, which must be set to the same value. **Table 3-3** lists the frequency bands supported by current products. For frequency bands defined in 3GPP specifications, see section 5.2 "Operating bands" in 3GPP TS 38.104 V18.6.0.

Table 3-3 Frequency bands supported by current products

Frequency Band	Uplink Frequency Range	Downlink Frequency Range	Duplex Mode
n38	2570 MHz to 2620 MHz	2570 MHz to 2620 MHz	TDD
n40	2300 MHz to 2400 MHz	2300 MHz to 2400 MHz	TDD
n41	2496 MHz to 2690 MHz	2496 MHz to 2690 MHz	TDD
n77	3300 MHz to 4200 MHz	3300 MHz to 4200 MHz	TDD
n78	3300 MHz to 3800 MHz	3300 MHz to 3800 MHz	TDD
n79	4400 MHz to 5000 MHz	4400 MHz to 5000 MHz	TDD
n80	1710 MHz to 1785 MHz	N/A	SUL
n82	832 MHz to 862 MHz	N/A	SUL
n83	703 MHz to 748 MHz	N/A	SUL
n84	1920 MHz to 1980 MHz	N/A	SUL
n86	1710 MHz to 1780 MHz	N/A	SUL
n257	26500 MHz to 29500 MHz	26500 MHz to 29500 MHz	TDD
n258	24250 MHz to 27500 MHz	24250 MHz to 27500 MHz	TDD
n261	27500 MHz to 28350 MHz	27500 MHz to 28350 MHz	TDD

3.1.3.2 Duplex Mode

The duplex mode of a cell is specified by the **NRDUCell.DuplexMode** parameter.

- If this parameter is set to **CELL_TDD**, the cell is a TDD cell.
- If this parameter is set to **CELL_SUL**, the cell is an SUL cell.

3.1.3.3 Cell Bandwidth

The uplink and downlink bandwidths of a cell are specified by the **NRDUCell.UlBandwidth** and **NRDUCell.DlBandwidth** parameters, respectively. For details about the bandwidths supported by an NR cell, see *Scalable Bandwidth*. **Table 3-4** describes the bandwidth requirements of different types of cells.

Table 3-4 Bandwidth requirements of different types of cells

Cell Duplex Mode	NRDUCell. <i>UL Bandwidth</i> (Uplink Bandwidth)	NRDUCell. <i>DL Bandwidth</i> (Downlink Bandwidth)	Remarks
TDD	Set this parameter as required.	Set this parameter as required.	The uplink and downlink bandwidths must be the same.
SUL	Set this parameter as required.	Set this parameter to NOT_CONFIG .	For details about the bandwidths supported by an SUL cell, see <i>UL and DL Decoupling</i> .

3.1.3.4 NR-ARFCN

The NR-ARFCN, center frequency of the cell carrier, is used to determine the center frequency-domain position of an NR DU cell.

NR-ARFCN Configuration Requirements

An NR-ARFCN can be an uplink NR-ARFCN or a downlink NR-ARFCN, which is specified by the **NRDUCell.*ULNarfcn*** or **NRDUCell.*DLNarfcn*** parameter, respectively. The NR-ARFCN configuration requirements of different cells are described as follows:

- For a TDD cell, the downlink NR-ARFCN must be configured and the uplink NR-ARFCN configuration is optional. If the uplink NR-ARFCN is not configured, it is considered to be the same as the downlink NR-ARFCN.
- For an SUL cell, the uplink NR-ARFCN must be configured and the downlink NR-ARFCN does not need to be configured.

NR-ARFCN Configuration Methods

The uplink and downlink NR-ARFCNs are calculated based on the planned start and end frequencies of a cell. The calculation procedure is as follows ([Table 3-5](#) lists the involved parameters):

- Determine the downlink center frequency of a cell based on the downlink start frequency and downlink bandwidth of the cell.

$$\text{Cell downlink center frequency (DLCELL}_{\text{REF}}) = \text{Cell downlink start frequency (MHz)} + \text{Cell downlink bandwidth (MHz)} / 2$$
- Determine the uplink center frequency of a cell based on the uplink start frequency and uplink bandwidth of the cell.

$$\text{Cell uplink center frequency (ULCELL}_{\text{REF}}) = \text{Cell uplink start frequency (MHz)} + \text{Cell uplink bandwidth (MHz)} / 2$$
- Calculate the downlink and uplink NR-ARFCNs of the cell based on the cell downlink center frequency (DLCELL_{REF}) and cell uplink center frequency (ULCELL_{REF}), respectively.

Downlink NR-ARFCN ($N_{\text{REF-DLCELL}}$) = Rounding up/down the result of $[(\text{DLCELL}_{\text{REF}} - F_{\text{REF-Offs}})/\Delta F_{\text{Global}} + N_{\text{REF-Offs}}]$ to the nearest integer as required in [Table 3-6](#)

Uplink NR-ARFCN ($N_{\text{REF-ULCELL}}$) = Rounding up/down the result of $[(\text{ULCELL}_{\text{REF}} - F_{\text{REF-Offs}})/\Delta F_{\text{Global}} + N_{\text{REF-Offs}}]$ to the nearest integer as required in [Table 3-6](#)

Table 3-5 Parameters related to $N_{\text{REF-CELL}}$

Frequency Range (MHz)	ΔF_{Global} (MHz)	$F_{\text{REF-Offs}}$ (MHz)	$N_{\text{REF-Offs}}$
0 to 3000	0.005	0	0
3000 to 24250	0.015	3000	600000
24250 to 100000	0.06	24250.08	2016667

Table 3-6 Requirements for available NR-ARFCNs

Frequency Band	Subcarrier Spacing	Available NR-ARFCN Requirement
n38	30 kHz	The NR-ARFCN must be an integer multiple of 20.
n40	30 kHz	The NR-ARFCN must be an integer multiple of 20.
n41	30 kHz	The NR-ARFCN must be an integer multiple of 6.
n77	30 kHz	The NR-ARFCN must be an even number.
n78	30 kHz	The NR-ARFCN must be an even number.
n79	30 kHz	The NR-ARFCN must be an even number.
n80	15 kHz	The NR-ARFCN must be an integer multiple of 20.
n82	15 kHz	The NR-ARFCN must be an integer multiple of 20.
n83	15 kHz	The NR-ARFCN must be an integer multiple of 20.
n84	15 kHz	The NR-ARFCN must be an integer multiple of 20.
n86	15 kHz	The NR-ARFCN must be an integer multiple of 20.

Frequency Band	Subcarrier Spacing	Available NR-ARFCN Requirement
n257	120 kHz	The NR-ARFCN must be an odd number.
n258	120 kHz	The NR-ARFCN must be an odd number.
n261	120 kHz	The NR-ARFCN must be an odd number.

3.1.3.5 SSB Frequency-Domain Position

In an NR system, a synchronization signal and PBCH block (SSB) carries the primary synchronization signal (PSS), secondary synchronization signal (SSS), and physical broadcast channel (PBCH). Each SSB consists of 240 contiguous subcarriers. A UE searches for cells based on received PSS and SSS signals and obtains necessary information from the received broadcast system information. The UE then selects the most appropriate cell to camp on using the received information.

An SSB frequency-domain position and how it is described are specified by the **NRDUCell.SsbFreqPos** and **NRDUCell.SsbDescMethod** parameters, respectively. The following two description methods can be used:

- **NR-ARFCN**
If the **NRDUCell.SsbDescMethod** parameter is set to **SSB_DESC_TYPE_NARFCN**, the **NRDUCell.SsbFreqPos** parameter indicates the NR-ARFCN corresponding to the SSB center frequency.
- **GSCN**
If the **NRDUCell.SsbDescMethod** parameter is set to **SSB_DESC_TYPE_GSCN**, the **NRDUCell.SsbFreqPos** parameter indicates the GSCN corresponding to the SSB center frequency.

NOTE

In SA networking or NSA and SA hybrid networking, the **NRDUCell.SsbDescMethod** parameter must be set to **SSB_DESC_TYPE_GSCN**.

In NSA networking, the **NRDUCell.SsbDescMethod** parameter must be set to **SSB_DESC_TYPE_GSCN** or **SSB_DESC_TYPE_NARFCN**.

When configuring the SSB frequency-domain position, ensure that all subcarriers of the SSB are within the carrier bandwidth (excluding the guard bandwidth) of a cell. Otherwise, the cell cannot be set up or work properly.

Table 3-7 lists the mapping between GSCNs and frequencies. For details, see section 5.4.3.1 "Synchronization raster and numbering" in 3GPP TS 38.104 V15.4.0.

Table 3-7 Mapping between GSCNs and frequencies

Frequency Range	SSB Frequency-Domain Position (SS _{REF})	Calculation Formula for GSCN	GSCN Range
0–3000 MHz	$N \times 1200 \text{ kHz} + M \times 50 \text{ kHz}$ N = 1–2499. The value of M can be 1, 3, or 5.	$3N + (M - 3)/2$	2–7498
3000–24250 MHz	$3000 \text{ MHz} + N \times 1.44 \text{ MHz}$ N = 0–14756	$7499 + N$	7499–22255
24250–100000 MHz	$24250.08 \text{ MHz} + N \times 17.28 \text{ MHz}$ N = 0–4383	$22256 + N$	22256–26639

The GSCN configuration must meet the requirements in section 5.4.3.3 "Synchronization raster entries for each operating band" in 3GPP TS 38.104 V15.6.0. For example, for band n41, the GSCN must be an integer multiple of 3; for band n79, the GSCN must be an integer multiple of 16.

 NOTE

When a cell is interfered in the downlink, it is recommended that the SSB frequency-domain position be configured in the non-interference range, so that the frequency-domain position of CORESET 0 does not fall into the interference range. This ensures that initial access is not affected by interference. For details about CORESET 0, see section 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V15.5.0.

3.1.3.6 Subcarrier Spacing

The subcarrier spacing of a cell is specified by the **NRDUCell.SubcarrierSpacing** parameter, and the related configuration requirements are described in [Table 3-8](#).

Table 3-8 Subcarrier spacing configuration requirements

Cell Duplex Mode	Subcarrier Spacing Configuration Requirement
TDD	30 kHz in low frequency bands and 120 kHz in high frequency bands
SUL	15 kHz

The SSB subcarrier spacing of a cell is specified by the **NRDUCell.SsbSubcarrierSpacing** parameter, and the related configuration requirements are described in [Table 3-9](#).

Table 3-9 SSB subcarrier spacing configuration requirements

Cell Duplex Mode	SSB Subcarrier Spacing Configuration Requirement
TDD	30 kHz in low frequency bands and 120 kHz or 240 kHz in high frequency bands

 NOTE

For cells in a high-frequency multi-carrier sector, the SSB subcarrier spacing configurations must be the same.

During network configuration, the SSB subcarrier spacing configurations of all high-frequency cells in the high-frequency network deployment area must be the same. This prevents the hybrid networking of high-frequency cells with different SSB subcarrier spacings.

In high-frequency TDD, the 240 kHz SSB subcarrier spacing configuration is a trial function.

 NOTE

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

3.1.3.7 Uplink-Downlink Slot Configuration

The uplink-downlink slot configuration must be set for TDD cells, which is specified by the **NRDUCell.SlotAssignment** and **NRDUCell.SlotStructure** parameters. For details, see *Standards Compliance*.

3.1.3.8 Cell Frame Offset

The cell frame offset is the offset of the start frame against a reference clock source for a cell. The unit of the frame offset is T_s , which equals 1/30720 ms. Frame offset can be expressed in the unit of ms as long as the value in T_s is divided by 30720.

The frame offset of an NR cell can be configured using the **gNBFreqBandConfig.FrameOffset** or **gNodeBParam.FrameOffset** parameter. The

value of **gNBFreqBandConfig.FrameOffset** takes precedence. If this parameter is either not configured or set to **4294967295**, the value of **gNodeBParam.FrameOffset** applies.

Requirements on Cell Frame Offset Configuration

When configuring the cell frame offset, note the following:

- Frame offsets must be the same for co-BBP or co-RF NR cells. Otherwise, some cells will fail to be activated.
- There are no requirements on the frame offsets of LTE and NR cells that share the same BBP.
- Frame offsets must be the same for LTE and NR cells that share the same RF module, except in the following special case: When NR cells with a slot configuration of single-period 8:2 (8_2_DDDDDDDSUU) and LTE cells share the same RF modules, the NR cell frame offsets must be 3 ms later than the LTE cell frame offsets. Otherwise, some cells will fail to be activated. For details, see section "Slot Configuration (TDD)" in *Standards Compliance*.

NOTE

For details about the LTE cell frame offset, see *Cell Management* in *eRAN Feature Documentation*.

3.1.3.9 Logical Root Sequence Index

Before a cell preamble sequence can be generated based on a physical root sequence and cyclic shift, a logical root sequence is required to search for the physical root sequence. The index to a logical root sequence is specified by the **NRDUCell.LogicalRootSequenceIndex** parameter. For details about the configuration requirements, see *gNodeBFunction Parameter Reference*, as described in [4 Parameters](#).

3.1.3.10 Transmit/Receive Mode

The transmit/receive mode of a cell is specified by the **NRDUCellTrp.TxRxMode** parameter. [Table 3-10](#) lists the transmit/receive modes supported in NR cells. For details about the supported frequency bands and cell bandwidths for each transmit/receive mode, see *gNodeBFunction Parameter Reference*, as described in [4 Parameters](#).

Table 3-10 Transmit/Receive modes supported in NR cells served by different base stations

Cell Duplex Mode	Cell Transmit/Receive Mode Supported by Macro Base Station	Cell Transmit/Receive Mode Supported by LampSite Base Station
TDD	Low-frequency TDD: 2T2R, 4T4R, 8T8R, 32T32R, and 64T64R High-frequency TDD: 2T2R, 4T4R, and 8T8R	1T1R, 2T2R, and 4T4R

Cell Duplex Mode	Cell Transmit/Receive Mode Supported by Macro Base Station	Cell Transmit/Receive Mode Supported by LampSite Base Station
SUL	2R and 4R, as SUL cells are limited to uplink reception	2R, as SUL cells are limited to uplink reception

 NOTE

The cell must have the same transmit/receive mode as the AAU or RRU. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

3.1.3.11 Cell Power

In terms of power configuration, the maximum transmit power of a single antenna ($P_{\max_out}$) is used to configure the transmit power of a low-frequency cell, while the maximum EIRP ($EIRP_{Cell}$) is used to configure the power of a high-frequency cell. The power configuration mode of a cell is specified by the **NRDUCellTrp.PowerConfigMode** parameter, and the formulas for calculating the cell power in different power configuration modes are described in [Table 3-11](#).

Table 3-11 Formulas for calculating the cell power in different power configuration modes

Power Configuration Mode	Parameter Value	Maximum Transmit Power of a Single Antenna ($P_{\max_out}$)	Total Transmit Power of a Cell (P_{Cell})
Power configuration in the unit of 0.1 dBm for low-frequency cells.	TRANSMIT_POWER	$P_{\max_out} \text{ (dBm)} = \text{NRDUCellTrp.MaxTransmitPower}$	$P_{Cell} \text{ (dBm)} = P_{\max_out} \text{ (dBm)} + 10 \times \lg(N_{Tx})$
Power configuration in the unit of mW for low-frequency cells.	TRANSMIT_POWER_MW	$P_{\max_out} \text{ (mW)} = \text{NRDUCellTrp.MaxTransmitPowerMw}$	$P_{Cell} \text{ (mW)} = P_{\max_out} \text{ (mW)} \times N_{Tx}$

Power Configuration Mode	Parameter Value	Maximum Transmit Power of a Single Antenna ($P_{\max_out}$)	Total Transmit Power of a Cell (P_{Cell})
Power configuration using the MaxEirp parameter for high-frequency cells.	EIRP	$P_{\max_out} \text{ (dBm)} = P_{Cell} \text{ (dBm)} - 10 \times \lg(N_{Tx})$	$P_{Cell} \text{ (dBm)} = EIRP_{Cell} \text{ (dBm)} - \text{Gain (dBi)}$ where $EIRP_{Cell} \text{ (dBm)} = \text{NRDUCellTrp.MaxEirp}$ - Gain (dBi) indicates the antenna gain of the RF module serving the cell.
$N_{Tx} = \text{NRDUCellTrp.TxRxMode}$ - If multiple cells are served by the same RRU or AAU, the sum of $P_{\max_out}$ values for these cells must be less than or equal to the maximum transmit power of the RRU or AAU. Otherwise, ALM-29870 NR DU Cell TRP Unavailable will be generated.			

 NOTE

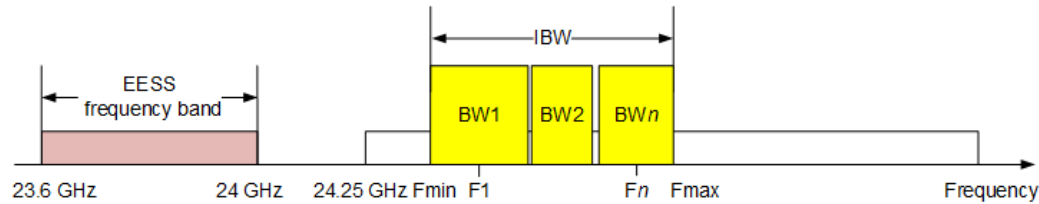
For details about the maximum transmit power of an RRU or AAU configured with different number of carriers, see *3900 & 5900 Series Base Station Product Documentation*.
SUL cells involve only uplink reception and therefore require no transmit power configuration.

Power Back-off Requirements of Cells Working in Band n258

To avoid interference of cells working in band n258 on Earth exploration-satellite service (EESS), a power back-off value must be configured for these high-frequency cells. (EESS frequency band spans from 23.6 GHz to 24 GHz.) The configuration requirements for the **NRDUCellTrp.MaxEirp** parameter are related to the power back-off values, which vary with the RF module. For details, see *Typical Carrier and Power Configurations for AAUs* in related technical specifications in *3900 & 5900 Series Base Station Product Documentation*. The following is an example of power back-off values.

Assume that a cell is served by the HAAU5213 (26 GHz), and the cell frequencies are $F_1, F_2, \dots$, and F_n (the corresponding cell bandwidths are $BW_1, BW_2, \dots$, and BW_n), as shown in [Figure 3-12](#). F_1 and F_n are the lowest and highest frequencies configured for the cell, respectively.

Figure 3-12 Interference avoidance for EESS



The power back-off value of the HAAU5213 is determined as follows:

- It is 3 dB when the following conditions are true: $IBW \leq 600$ MHz and $(2 \times F_{min} - F_{max}) < 24$ GHz.
- It is 5 dB when the following conditions are true: $IBW > 600$ MHz and $(2 \times F_{min} - F_{max}) < 24$ GHz.

where

- F_{min} (lower boundary of F_1) = $F_1 - BW/2$
- F_{max} (upper boundary of F_n) = $F_n + BW/2$
- IBW (instantaneous bandwidth) = $F_{max} - F_{min}$

3.1.3.12 Cell Service Type Tag

During large-scale parameter inspection, check, and O&M, the **NRCelloamParam.ServiceTypeTag** parameter is used to configure the cell service type tag to help quickly distinguish ToB and ToC service types.

The **NRCelloamParam.ServiceTypeTag** parameter can be set to **ToB**, **ToC**, **ToB_ToC**, or **NOT_CONFIG**. The default value is **NOT_CONFIG**. It is recommended that this parameter be set based on the network plan.

NOTE

The **NRCelloamParam.ServiceTypeTag** parameter indicates the tag of service types for a cell, which is used only for O&M and does not affect any base station functions.

3.2 Network Analysis

3.2.1 Benefits

The primary function of gNodeBs is to provide radio cell coverage for UEs. As such, cells must be activated first.

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

When unbalanced downlink transmit power configurations between channels are used:

- The **NRDUCellTrp.MaxTransmitPower** or **NRDUCellTrp.MaxTransmitPowerMw** parameter must be set to the average transmit power of all channels first. Otherwise, the calculated total cell power is inaccurate, causing UE throughput to decrease.
- If the power difference between channels is greater than 3 dB, beam exceptions may occur, causing UE throughput to decrease.

In scenarios where multiple UEs are performing downlink full buffer services in a cell set up on the UMPTe, the UE throughput cannot reach the optimal value due to the UMPTe performance limitations. Therefore, it is recommended that the UMPTg or a later main control board be used.

3.3 Requirements

3.3.1 Licenses

Cell establishment requires capacity licenses. For details about capacity licenses, see *License Control Item Lists*.

Establishment of SUL cells also requires feature licenses. For details, see license requirements in *UL and DL Decoupling*.

Establishment of hyper cells also requires feature licenses. For details, see license requirements in *Hyper Cell*.

Establishment of combined cells also requires feature licenses. For details, see license requirements in *Cell Combination*.

Establishment of distributed massive MIMO cells also requires feature licenses. For details, see license requirements in *Distributed Antenna Solutions (Low-Frequency TDD)*.

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

None

3.3.2.2 Mutually Exclusive Functions

Some features or functions are not supported in certain cell scenarios. For details about the cell requirements of these features or functions, see section "Requirements" in the corresponding feature documents.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	None	None	None	None

3.3.2.3 Function Impacts

Whether some features or functions take effect as expected vary with the cell scenario. For details about the cell requirements of these features or functions, see section "Requirements" in the corresponding feature documents.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Static GTP-U echo	None	<i>IPv4 Transmission</i> <i>IPv6 Transmission</i>	Static GTP-U echo must be enabled before the S1_U_OM_FAULT_AUTO_DEA_SW or COMPT_S1_U_UNAVAIL_AUTO_DEA_SW option of the gNodeBParam.CellStatePolicy-Switch parameter is selected.
Low-frequency TDD	Air-interface-based network-wide synchronization deviation detection	GAP_BASED_SYNC_DIFF_DETECT_SW option of the gNodeBParam.SyncDiffDetectSwitch parameter	<i>Synchronization</i>	Air-interface-based network-wide synchronization deviation detection does not take effect in combined-RRU cells.

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

For the hardware requirements (base station models, boards, RF modules, and other basic requirements) for establishing a common cell, see [Base Station Models, Boards, RF Modules](#), and [Others](#), respectively.

To establish a cell in other typical scenarios, the hardware requirements for establishing a common cell must be met. The hardware requirements for each scenario are as follows:

- For cells in a multi-carrier sector, see [Cells in a Multi-Carrier Sector](#).
- For SUL cells, see *UL and DL Decoupling*.
- For combined-RRU cells, see [Combined-RRU Cells](#).
- For split-sector cells, see [Split-Sector Cells](#).
- For RF combination cells, see [RF Combination Cells](#).
- For hyper cells, see *Hyper Cell*.
- For combined cells, see *Cell Combination*.
- For distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

- For supplementary cells, see [Supplementary Cells](#).

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Others

Other basic requirements for setting up cells:

- The hardware resources of BBPs, CPRI ports, and RF modules meet the cell requirements. For the hardware requirements of macro base stations, see related technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- The transmit/receive mode of the cell must match the number of transmit/receive channels of the RF module. Otherwise, the cell will not work normally. For details about the number of transmit/receive channels supported by RF modules, see related technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- The CPRI data rates for the optical modules must match the cell bandwidths and transmit/receive modes when selecting optical modules for BBPs. For details about the constraints between the CPRI data rates and the cell bandwidths and transmit/receive modes, see related technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- Cells served by the RRUs/AAs must work in frequency bands supported by the RRUs/AAs.
- 32T32R, 16T16R, and 64T64R cells must be established by using AAs.
- Each eight-antenna cell must be established by using an 8T8R RRU/AA.
- For details about the distance between the BBU and RF modules, see *RRU Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*. Moreover, the two optical fibers used in the load sharing or ring topology must be the same length.
- In the load sharing topology, if two optical fibers connect to two different BBPs, neither of the BBPs can be a UBBPfw. It is recommended that the optical module rates be the same for the two optical fibers. Otherwise, cell setup may be affected.

- It is recommended that low-frequency cells be deployed on the BBPs directly connecting to the CPRI/eCPRI optical fibers. If inter-BBP deployment is required, it is recommended that the cells be bound to the BBPs to reduce the time required for cell setup attempts.
- Two or more cells with overlapping spectrum resources (other than guard bands) cannot be configured on the same RRU/AAU antenna channel.
- A calibration cable must be connected to the calibration port of an integrated 8T8R RRU. The function of cable connection check for the calibration port on an 8T RRU (controlled by the **RRU_CALIBPORT_CONN_CHK_SW** option of the **NRDUCellChnCalib.CalibrationAlgoSwitch** parameter) can be used to check the cable connection status. After this function is enabled, **ALM-29871 NR DU Cell TRP Capability Degraded** with **Specific Problem** being **Low signal strength of calibration channel** will be reported when the connection status of a calibration cable becomes abnormal. The **RRU.CALPORTCONSTATUS** parameter must be set to **CONNECTED** before a cable connection check. All NR TDD-capable 8T8R RRU5XXX modules except the RRU5254, RRU5258, and RRU5259 support this function. (X indicates a digit.) This function requires that the **NRDUCellTrp.TxRxMode** parameter be set to **8T8R**. This function is not supported when two RRUs are combined to serve an 8T8R cell.

Extra requirements for configuring NR TDD cells:

- If NR TDD cells served by the same base station have different slot configurations, they must be bound to different BBPs. This is because the slot configurations of all NR TDD cells on the same BPB must be the same.
- A cell will fail to be set up due to frame configuration conflicts in either of the following scenarios:
 - The slot configurations or slot structures are inconsistent among multiple NR carriers.
 - The uplink and downlink slot configurations for software-defined radio (SDR) RF modules are different between LTE and NR. For details, see *Standards Compliance*.
- If NR TDD cells under a base station have different subcarrier spacing, they must be bound to different BBPs. The subcarrier spacing of all NR TDD cells on the same BPB must be the same.
- When multiple cells, which consist of only NR cells or both NR and LTE cells, are to be set up on the same RF module, the TA offset (specified by the **NRDUCell.TaOffset** parameter) must meet the following requirements. Otherwise, subsequent cell setup will fail among these cells or the network performance of all cells served by this RF module will deteriorate.
 - For the NR cells served by this RF module, the TA offsets must be the same.
 - For the NR and LTE cells served by this RF module, the TA offset of the NR cells must be the same as that of the LTE cells, which is 39936TC.
- For details about bandwidth and transmit/receive mode requirements of one BPB, see related technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- If both eCPRI 64T64R NR TDD cells and CPRI 64T64R NR TDD cells are served by the same base station, they must be bound to different BBPs.

- Extra requirements for configuring high-frequency TDD cells:
 - High-frequency cells must be established on the BBPs directly connecting to the CPRI/eCPRI optical fibers, and such cells must also be bound to these BBPs, if required.
 - For high-frequency cells, CPRI deployment and eCPRI deployment cannot be implemented on the same BBP.

Extra requirements for configuring cells served by AAUs:

- For cells served by AAUs using eCPRI ports, the number of single-RAT carriers cannot exceed 10 and the number of multi-RAT carriers cannot exceed 16.
- For cells served by AAUs using CPRI ports, the total number of carriers (regardless of single-RAT or multi-RAT) cannot exceed 16.

Cells in a Multi-Carrier Sector

Requirements for setting up cells in a multi-carrier sector:

- The RRUs/AAUs/pRRUs must support multiple carriers. For details about the RRUs/AAUs/pRRUs supporting multiple carriers, see related technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- The frequency range of each cell must be within the frequency bands supported by the RRUs/AAUs/pRRUs.
- When multiple cells are served by the same RF module, the slot configuration and slot structure settings of these cells must be the same. When multiple cells in different NR frequency bands are served by an AAU5665md, these cells can use different slot configuration and slot structure settings. When multiple cells in different NR frequency bands are served by an AAU5368md, these cells can use different slot configuration and slot structure settings but must use the same frame offset. When multiple cells in different NR frequency bands are served by the same RF module among the pRRU565xAB/pRRU575xAB/pRRU585xAB series modules (*x* indicates a digit, and *A* and *B* indicate different letters), these cells can use different slot configuration and slot structure settings.
- The frequency range of each cell cannot overlap with one another.
- In high frequency bands, for cells in a multi-carrier sector, a single RF module can be configured for only one sector and one piece of sector equipment. However, multiple RF modules cannot be configured for the same sector.
- In high frequency bands, for cells in a multi-carrier sector, certain parameters of each cell in the same sector must be set to the same value. Otherwise, network performance may be affected. For details about related parameters, see [3.4.1.1 Data Preparation](#).
- When a low-frequency TDD massive MIMO RF module that supports an RET system serves multiple cells, the tilt and coverage scenario configurations of these cells must meet certain conditions. Incorrect configurations may affect cell performance. For details, see *Beam Management*.

Combined-RRU Cells

The requirements of combined-RRU cells on base station models are as follows:

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Other requirements of combined-RRU cells are as follows:

- The two RRUs must support the same frequency range and have the same number of transmit and receive antennas.
- Two 2R RRUs of the same model can be combined to serve a 4R SUL cell.
- Two 2T2R RRUs of the same model or two 2T4R RRUs of the same model can be combined to serve a 4T4R cell.
- The length difference between the CPRI optical fibers used to connect the two RRUs to the BBU cannot exceed 2 km. Otherwise, the cell may be unavailable.
- The CPRI optical fibers of two RRUs must connect to the BBPs of the same RAT. For details, see *CPRI MUX*.
- Only a star or cascading topology can be used for connecting a BBU and two RRUs. If RRUs are cascaded, the distance between the BBU and RRUs must not exceed 20 km. For details, see *RF Unit and Topology Management*.
- If a combination of two RRUs in a star topology applies to three-sector SUL cells, six CPRI ports are required. If the BBP cannot provide enough CPRI ports, cascading networking (where the optical fiber length between two RRUs to be cascaded is less than 100 m) or LNR CPRI MUX networking can be used. For details, see *RF Unit and Topology Management*. Two combined RRUs must connect to the same BBP.

Split-Sector Cells

Requirements of split-sector cells on base station models: 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

If a 4T4R RRU is to be connected to two antennas to serve two split-sector 2T2R cells, all NR-capable 4T4R RF modules can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

RF Combination Cells

A pRRU group can consist of 5000 series pRRUs but cannot consist of both 3000 series and 5000 series pRRUs.

If the pRRU5551JR/pRRU565xAB/pRRU575xAB/pRRU585xAB modules are used together with pRRUs of other models, RF combination cells cannot be established. (*x* is a digit, and *A* and *B* stand for different letters.)

Supplementary Cells

The UBBPg3/UBBPh3a/UBBPh3b must be used.

Requirements on RF modules:

- For macro base stations, all of the NR TDD-capable AAUs that work in high frequency bands support this function. For details, see the technical specifications of AAUs in *3900 & 5900 Series Base Station Product Documentation*.

- For LampSite base stations, the pRRU5551JR supports this function. For details, see *LampSite pRRU Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*.

3.3.4 Others

None

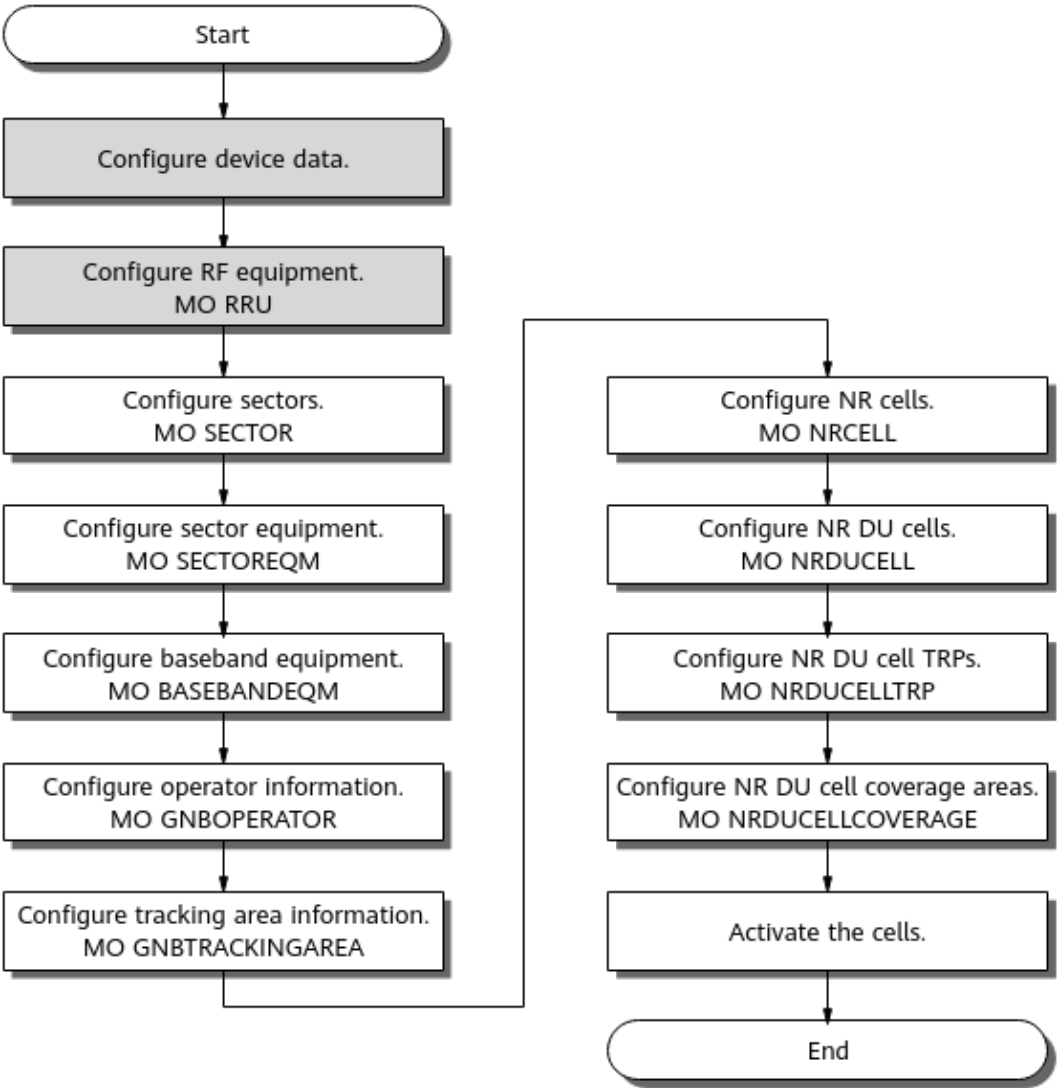
3.4 Operation and Maintenance

3.4.1 Data Configuration

3.4.1.1 Data Preparation

Figure 3-13 shows the cell deployment process.

Figure 3-13 Cell deployment process



 NOTE

For details about how to configure device data, see *gNodeB Initial Configuration Guide (MML-based)* in *3900 & 5900 Series Base Station Product Documentation*.

For details about the parameters and MML command examples for configuring RF equipment, see *RF Unit and Topology Management*.

The following table describes the parameters that must be set in a **SECTOR** MO to configure a sector.

Parameter Name	Parameter ID	Setting Notes
Sector ID	SECTOR.SECTORID	Set this parameter to a unique value at the base station level.
Sector Name	SECTOR.SECNAME	Set this parameter based on the network plan.
Location Name	SECTOR.LOCATIONNAME	Set this parameter based on the network plan.
User Label	SECTOR.USERLABEL	Set this parameter based on the network plan.
Create Default Sector Equipment	SECTOR.CREATESECTOREQM	Set this parameter based on the network plan.
Antenna Number	SECTOR.ANTNUM	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **SECTOREQM** MO to configure a piece of sector equipment.

Parameter Name	Parameter ID	Setting Notes
Sector Equipment ID	SECTOREQM.SECTOREQMID	Set this parameter based on the network plan.
Sector ID	SECTOREQM.SECTORID	Set this parameter based on the network plan.
Sector Equipment Antenna	SECTOREQM.SECTOREQMANTENNA	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Antenna Config Mode	SECTOREQM.ANTCFG MODE	In low frequency bands: - When the NRDUCellTrp.TxRxMode parameter is set to 32T32R, 16T16R, or 64T64R for an NR DU cell TRP, set this parameter to BEAM. - For any other transmit/receive mode, set this parameter to the default value ANTENNAPORT. In high frequency bands, set this parameter to BEAM.
RRU Cabinet No.	SECTOREQM.RRUCN	Set this parameter based on the network plan.
RRU Subrack No.	SECTOREQM.RRUSRN	Set this parameter based on the network plan.
RRU Slot No.	SECTOREQM.RRUSN	Set this parameter based on the network plan.
Beam Shape	SECTOREQM.BEAMSH APE	This parameter is valid and is set to SEC_120DEG when the SECTOREQM.ANTCFGMODE parameter is set to BEAM .
Beam Layer Split	SECTOREQM.BEAMLA YERSPLIT	This parameter is valid and is set to None when the SECTOREQM.ANTCFGMODE parameter is set to BEAM .
Beam Azimuth Offset	SECTOREQM.BEAMAZ IMUTHOFFSET	This parameter is valid and is set to None when the SECTOREQM.ANTCFGMODE parameter is set to BEAM .
Antenna Number	SECTOREQM.ANTNUM	Set this parameter based on the network plan.
Cabinet No. of Antenna 1	SECTOREQM.ANT1CN	This parameter is mandatory when the SECTOREQM.ANTCFGMODE parameter is set to ANTENNAPORT .
Subrack No. of Antenna 1	SECTOREQM.ANT1SR N	This parameter is mandatory when the SECTOREQM.ANTCFGMODE parameter is set to ANTENNAPORT .

Parameter Name	Parameter ID	Setting Notes
Slot No. of Antenna 1	SECTOREQM.ANT1SN	This parameter is mandatory when the SECTOREQM.ANTCFGMODE parameter is set to ANTENNAPORT .
Channel No. of Antenna 1	SECTOREQM.ANT1N	This parameter is mandatory when the SECTOREQM.ANTCFGMODE parameter is set to ANTENNAPORT .
Antenna 1 RX/TX Mode	SECTOREQM.ANTTYP E1	This parameter is mandatory when the SECTOREQM.ANTCFGMODE parameter is set to ANTENNAPORT .

For details about how to configure parameters related to RET antennas and RET subunits in both new and old models, see *ALD Management in 3900 & 5900 Series Base Station Product Documentation*.

The following table describes the parameters that must be set in a **BASEBANDEQM** MO to configure baseband equipment information.

Parameter Name	Parameter ID	Setting Notes
Baseband Equipment ID	BASEBANDEQM.BASE BANDEQMID	Each BASEBANDEQMID parameter value identifies a piece of baseband equipment. One piece of baseband equipment can include multiple BBPs.
Baseband Equipment Type	BASEBANDEQM.BASE BANDEQMTYPE	Set this parameter based on the network plan.
Baseband Equipment Board	BASEBANDEQM.BASE BANDEQMBOARD	Set this parameter based on the network plan.
UMTS UL Demodulation Mode	BASEBANDEQM.UMT SDEMMODE	It is recommended that this parameter be set to NULL because it is not used for gNodeBs.

The following table describes the parameters that must be set in a **gNBOperator** MO to configure an operator.

Parameter Name	Parameter ID	Setting Notes
Operator ID	gNBOperator.OperatorId	Set this parameter based on the network plan.
Operator Name	gNBOperator.OperatorName	Set this parameter based on the network plan.
Operator Type	gNBOperator.OperatorType	Set this parameter based on the network plan.
Mobile Country Code	gNBOperator.Mcc	Set this parameter based on the network plan.
Mobile Network Code	gNBOperator.Mnc	Set this parameter based on the network plan.
NR Networking Option	gNBOperator.NrNetworkingOption	- Set this parameter to SA or SA_NSA in SA networking. - Set this parameter to NSA or SA_NSA in NSA networking.

The following table describes the parameters that must be set in a **gNBTrackingArea** MO to configure a tracking area for an operator.

Parameter Name	Parameter ID	Setting Notes
Tracking Area ID	gNBTrackingArea.TrackingAreaId	The setting of this parameter takes effect only for the gNodeB. The tracking area identity (TAI) specified by this parameter is different from the TAI configured in the tracking area list (TAL) in the core network.
Tracking Area Code	gNBTrackingArea.Tac	- Tracking area codes (TACs) do not need to be planned for NSA cells. You are advised to set this parameter to an invalid value (4294967295). - TACs must be planned for SA cells and cannot be set to an invalid value.

The following table describes the parameters that must be set in an **NRCell** MO to configure an NR cell.

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCell.NrCellId	Set this parameter based on the network plan.
Cell Name	NRCell.CellName	Set this parameter based on the network plan.
Frequency Band	NRCell.FrequencyBand	Set this parameter based on the network plan.
Cell ID	NRCell.CellId	For details, see sections 9.3.1.6 "Global gNB ID" and 9.3.1.7 "NR CGI" in 3GPP TS 38.413 V15.4.0. The length of this parameter value is the length of NR Cell Identity (36 bits) minus the length of gNodeB ID (22 bits to 32 bits). For example, if the length of gNodeB ID is 22 bits, set this parameter to a value indicating 14 bits.
Duplex Mode	NRCell.DuplexMode	Set this parameter based on the network plan.
User Label	NRCell.UserLabel	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **NRDUCell** MO to configure an NR DU cell.

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCell.NrDuCellId	Set this parameter based on the network plan.
NR DU Cell Name	NRDUCell.NrDuCellName	Set this parameter based on the network plan.
Frequency Band	NRDUCell.FrequencyBand	Set this parameter based on the network plan.
Uplink Bandwidth	NRDUCell.UlBandwidth	Set this parameter based on the network plan.
Downlink Bandwidth	NRDUCell.DlBandwidth	Set this parameter based on the network plan.
Duplex Mode	NRDUCell.DuplexMode	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Logical Root Sequence Index	NRDUCell.LogicalRootSequenceIndex	Set this parameter based on the network plan.
PRACH Frequency Start Position	NRDUCell.PrachFrequencyStartPosition	Set this parameter based on the network plan.
Cell Radius	NRDUCell.CellRadius	The value of this parameter must be greater than or equal to the actual cell radius. Otherwise, UEs at the cell edge may fail to access the network.
Slot Assignment	NRDUCell.SlotAssignment	This parameter needs to be set only for TDD cells.
Slot Structure	NRDUCell.SlotStructure	This parameter needs to be set only for TDD cells.
Cell ID	NRDUCell.CellId	The value of this parameter must be the same as that of the NRCell.CellId parameter.
Uplink NARFCN	NRDUCell.UlNarfcn	Set this parameter based on the network plan. For details, see 3.1.3.4 NR-ARFCN .
Downlink NARFCN	NRDUCell.DlNarfcn	Set this parameter based on the network plan. For details, see 3.1.3.4 NR-ARFCN .
Physical Cell ID	NRDUCell.PhysicalCellId	If the uplink or downlink NR-ARFCNs of two NR DU cells with overlapping parts are the same, the NRDUCell.PhysicalCellId settings of the two cells must be different in order to avoid mutual signal interference.
Subcarrier Spacing	NRDUCell.SubcarrierSpacing	Set this parameter based on the network plan.
Tracking Area ID	NRDUCell.TrackingAreaId	This parameter must be set to the same value as the Tracking Area ID parameter in the gNBTrackingArea MO.
SSB Frequency Position Describe Method	NRDUCell.SsbDescribeMethod	Set this parameter based on the network plan. For details, see 3.1.3.5 SSB Frequency-Domain Position .

Parameter Name	Parameter ID	Setting Notes
SSB Frequency Position	NRDUCell.SsbFreqPos	Set this parameter based on the network plan. For details, see 3.1.3.5 SSB Frequency-Domain Position .
Supplementary Cell Indicator	NRDUCell.SupplementaryCellInd	Set this parameter based on the network plan. For details, see 3.1.2.10 Supplementary Cell .
SSB Subcarrier Spacing	NRDUCell.SsbSubcarrierSpacing	Set this parameter based on the network plan. For details, see 3.1.3.6 Subcarrier Spacing .
NR DU Cell Networking Mode	NRDUCell.NrDuCellNetworkingMode	Set this parameter based on the network plan.
LampSite Cell Flag	NRDUCell.LampSiteCellFlag	Set this parameter based on the network plan.
Cell Administration State	NRDUCell.CellAdministrationState	Set this parameter based on the network plan.
Deployment Position	NRDUCell.DeploymentPosition	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **NRDUCellTrp** MO to configure the TRP for an NR DU cell.

Parameter Name	Parameter ID	Setting Notes
NR DU Cell TRP ID	NRDUCellTrp.NrDuCellTrpId	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCellTrp.NrDuCellId	Set this parameter based on the network plan.
Transmit and Receive Mode	NRDUCellTrp.TxRxMode	Set this parameter based on the network plan.
Baseband Equipment ID	NRDUCellTrp.BasebandEqmId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Power Config Mode	NRDUCellTrp.PowerConfigMode	Set this parameter to TRANSMIT_POWER or TRANSMIT_POWER_MW for low-frequency cells and to EIRP for high-frequency cells. This parameter does not take effect when the NRDUCellTrp.TxRxMode parameter is set to 2R or 4R .
Max Transmit Power	NRDUCellTrp.MaxTransmitPower	Set this parameter based on the network plan. This parameter is valid and mandatory only when the NRDUCellTrp.PowerConfigMode parameter is set to TRANSMIT_POWER .
Max Transmit Power in Milliwatt	NRDUCellTrp.MaxTransmitPowerMw	Set this parameter based on the network plan. This parameter is valid and mandatory only when the NRDUCellTrp.PowerConfigMode parameter is set to TRANSMIT_POWER_MW .
Max EIRP	NRDUCellTrp.MaxEirp	Set this parameter based on the network plan. This parameter is valid and mandatory only when the NRDUCellTrp.PowerConfigMode parameter is set to EIRP .
CPRI Compression	NRDUCellTrp.CpriCompression	Set this parameter based on the network plan.
Frequency Range and Duplex Mode	NRDUCellTrp.FrAndDuplexMode	Set this parameter based on the network plan. The TRP beam configuration of a cell can be queried based on the value of this parameter.

The following table describes the parameters that must be set in an **NRDUCellCoverage** MO to configure information related to the coverage area of an NR DU cell.

Parameter Name	Parameter ID	Setting Notes
NR DU Cell TRP ID	NRDUCellCoverage.NrDuCellTrpld	Set this parameter based on the network plan.
NR DU Cell Coverage ID	NRDUCellCoverage.NrDuCellCoverageId	Set this parameter based on the network plan.
Sector Equipment ID	NRDUCellCoverage.SectorEqmId	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **NRDUCellCoverage** MO to configure the offset of the transmit power of each physical channel relative to the reference transmit power when downlink transmit power configurations between channels are unbalanced.

Parameter Name	Parameter ID	Setting Notes
Transmit Power Offset of Channel 0	NRDUCellCoverage.TransmitPowerOffsetChannel0	Set this parameter based on the network plan.
Transmit Power Offset of Channel 1	NRDUCellCoverage.TransmitPowerOffsetChannel1	Set this parameter based on the network plan.
Transmit Power Offset of Channel 2	NRDUCellCoverage.TransmitPowerOffsetChannel2	Set this parameter based on the network plan.
Transmit Power Offset of Channel 3	NRDUCellCoverage.TransmitPowerOffsetChannel3	Set this parameter based on the network plan.

The following table describes the parameter that must be set in an **NRCelloamParam** MO to configure O&M information for an NR cell.

Parameter Name	Parameter ID	Setting Notes
Service Type Tag	NRCelloamParam.ServiceTypeTag	Set this parameter based on the network plan.

Precautions for Configuring Cells in a Multi-Carrier Sector in High Frequency Bands

In high frequency bands, all cells in a multi-carrier sector share one beam. Therefore, the configurations of these cells must be the same. That is, the settings

of all parameters in MOs whose unique primary key parameter is **NrDuCellId** or **NrDuCellTrpId** must be the same. However, the following parameters of basic cell attributes are not subject to this restriction.

- The following parameters in the **NRDUCell** MO:
 - Cell ID (**NRDUCell.CellId**)
 - NR DU Cell Name (**NRDUCell.NrDuCellName**)
 - Uplink NARFCN (**NRDUCell.UlNarfcn**)
 - Downlink NARFCN (**NRDUCell.DlNarfcn**)
 - SSB Frequency Position Describe Method (**NRDUCell.SsbDescMethod**)
 - SSB Frequency Position (**NRDUCell.SsbFreqPos**)
 - Frequency Band (**NRDUCell.FrequencyBand**)
 - Additional Frequency Band (**NRDUCell.AdditionalFrequencyBand**)
 - Supplementary Cell Indicator (**NRDUCell.SupplementaryCellInd**)
- The following parameter in the **NRDUCellTrp** MO:
 - NR DU Cell ID (**NRDUCellTrp.NrDuCellId**)

3.4.1.2 Using MML Commands

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met. This section provides MML command examples for establishing a cell. Parameter values in the following examples are for reference only. On live networks, these parameters must be configured based on the operator's network plan. For details about MML command examples for RF equipment, see *RF Unit and Topology Management*.

- For details about the MML command examples for hyper cells, see *Hyper Cell*.
- For details about MML command examples for combined cells, see *Cell Combination*.
- For details about MML command examples for distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD)*.

In LampSite scenarios, only [3.4.1.2.1 Establishing a TDD Cell](#) and [3.4.1.2.6 Establishing an RF Combination Cell](#) are supported.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Process

For details about the cell deployment process, see [Figure 3-13](#).

Assume that cells with different transmit/receive modes, duplex modes, or bandwidths are deployed on the same BBP. If ALM-29870 NR DU Cell TRP Unavailable with **Specific Problem** being "Redeployment required due to baseband resource conflicts" is reported due to incorrect BBP mode, run the **STR**

NRDUCELLREDEPLOY or **STR BRDBBRESREALLOC** command as a solution. Within 15 minutes after either command is executed, if NR DU cell TRPs are unavailable because redeployment is required due to baseband resource conflicts, automatic restart and redeployment will be triggered for the cells deployed on the corresponding baseband equipment.

```
//Automatically restarting and redeploying conflicting cells that cause cell TRP unavailability (This is a high-risk command.)
STR NRDUCELLREDEPLOY: DeployType=CONFLICT;
//Starting automatic reallocation of baseband resources (This is a high-risk command.)
STR BRDBBRESREALLOC: CN=0, SRN=0, SN=1;
```

3.4.1.2.1 Establishing a TDD Cell

Figure 3-13 shows the general process of establishing a common cell. To establish a cell in other scenarios, perform the general process with any necessary adjustments.

NOTE

- The transmit/receive mode of a cell must match the number of antennas in the sector serving the cell. The sector must also be configured with the required number of antennas (specified by the **SECTOR.ANTNUM** parameter). For example, if a cell works in 8T8R mode, the **SECTOR.ANTNUM** parameter must be set to 8.
- If the **SECTOR.CREATESECTOREQM** parameter under **ADD SECTOR** is set to **TRUE**, there is no need to run the **ADD SECTOREQM** command. The **SECTOR.CREATESECTOREQM** parameter is set to **FALSE** by default.

Establishing a 64T64R massive MIMO TDD cell (using a 100 MHz cell as an example)

```
//Adding a sector
ADD SECTOR: SECTORID=0, SECNAME="0", LOCATIONNAME="0", USERLABEL="0", ANTNUM=0,
CREATESECTOREQM=FALSE;
//Adding sector equipment, and setting SectorEqm.SectorId to specify the sector where the sector
equipment is located
ADD SECTOREQM: SECTOREQMID=0, SECTORID=0, ANTCFGMODE=BEAM, RRUCN=0, RRUSRN=60,
RRUSN=0, BEAMSHAPE=SEC_120DEG, BEAMLAYERSPLIT=None, BEAMAZIMUTHOFFSET=None;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding a 100 MHz TDD cell
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N78, DuplexMode=CELL_TDD,
UserLabel="CELL1";
//Adding an NR DU cell, and setting NRDUCell.CellId to associate it with the NR cell
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="CELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N78, DInarfcn=634000, UIBandwidth=CELL_BW_100M,
DIBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7853, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP, setting NRDUCellTrp.NrDuCellId to specify the NR DU cell served by this TRP,
and setting NRDUCellTrp.BasebandEqmId to bind baseband equipment to the NR DU cell TRP (If no
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell
TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=1, TxRxMode=64T64R,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=300, BasebandEqmId=0,
CpriCompression=3DOT2_COMPRESSION, FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, setting NRDUCellCoverage.NrDuCellTrpId to specify the NR DU cell
TRP corresponding to this coverage area, and setting NRDUCellCoverage.SectorEqmId to bind sector
```

```
equipment to the NR DU cell TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Establishing an 8T8R TDD cell (using a 100 MHz cell as an example)

```
//Adding a sector and sector equipment, with the SECTOR.ANTNUM parameter set to 8, and configuring
eight antenna channels of an RRU for this sector
ADD SECTOR: SECTORID=0, SECNAME="0", USERLABEL="0", ANTNUM=8, ANT1CN=0, ANT1SRN=60,
ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=60,
ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, ANT5CN=0, ANT5SRN=60,
ANT5SN=0, ANT5N=R0E, ANT6CN=0, ANT6SRN=60, ANT6SN=0, ANT6N=R0F, ANT7CN=0, ANT7SRN=60,
ANT7SN=0, ANT7N=R0G, ANT8CN=0, ANT8SRN=60, ANT8SN=0, ANT8N=R0H, CREATESECTOREQM=TRUE,
SECTOREQMID=0;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding a 100 MHz TDD cell
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N78, DuplexMode=CELL_TDD;
//Adding an NR DU cell, and setting NRDUCell.CellId to associate it with the NR cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N78, DfNarfcn=634000, UlBandwidth=CELL_BW_100M,
DlBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7853, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP, setting NRDUCellTrp.NrDuCellId to specify the NR DU cell served by this TRP,
and setting NRDUCellTrp.BasebandEqmId to bind baseband equipment to the NR DU cell TRP (If no
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell
TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=8T8R,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=320, BasebandEqmId=0,
CpriCompression=3DOT2_COMPRESSION, FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, setting NRDUCellCoverage.NrDuCellTrpId to specify the NR DU cell
TRP corresponding to this coverage area, and setting NRDUCellCoverage.SectorEqmId to bind sector
equipment to the NR DU cell TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;
//Activating the cell
ACT NRCELL: NrCellId=0;
```

Establishing a 4T4R TDD cell (using a 100 MHz cell as an example)

```
//Adding a sector and sector equipment, with the SECTOR.ANTNUM parameter set to 4, and four antenna
channels of an RRU configured for this sector
ADD SECTOR: SECTORID=0, SECNAME="0", USERLABEL="0", ANTNUM=4, ANT1CN=0, ANT1SRN=60,
ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=60,
ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,
SECTOREQMID=0;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding a 100 MHz TDD cell
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N78, DuplexMode=CELL_TDD;
//Adding an NR DU cell, and setting NRDUCell.CellId to associate it with the NR cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N78, DfNarfcn=634000, UlBandwidth=CELL_BW_100M,
DlBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
```

```
SsbFreqPos=7853, SsbSubcarrierSpacing=30KHZ;  
//Adding an NR DU cell TRP, setting NRDUCellTrp.NrDuCellId to specify the NR DU cell served by this TRP,  
and setting NRDUCellTrp.BasebandEqmId to bind baseband equipment to the NR DU cell TRP (If no  
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell  
TRP.)  
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=4T4R,  
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=320, BasebandEqmId=0,  
CpriCompression=3DOT2_COMPRESSION, FrAndDuplexMode=FR1_TDD;  
//Adding an NR DU cell coverage area, setting NRDUCellCoverage.NrDuCellTrpId to specify the NR DU cell  
TRP corresponding to this coverage area, and setting NRDUCellCoverage.SectorEqmId to bind sector  
equipment to the NR DU cell TRP  
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;  
ACT NRCELL: NrCellId=0;
```

Establishing a LampSite 4T4R TDD cell (using a 100 MHz cell as an example)

```
//Adding a sector and sector equipment, with the SECTOR.ANTNUM parameter set to 4  
ADD SECTOR: SECTORID=0, SECNAME="0", USERLABEL="0", ANTNUM=4, ANT1CN=0, ANT1SRN=60,  
ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=60,  
ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,  
SECTOREQMID=0;  
//Adding baseband equipment  
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=0;  
  
//(Optional) Adding operator and tracking area information (if not added previously)  
//Adding an operator "XXXX" and setting NR networking option as SA or NSA  
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",  
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA;  
//Adding a tracking area  
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;  
  
//Adding a 100 MHz TDD cell  
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N77, DuplexMode=CELL_TDD;  
//Adding an NR DU cell, and setting NRDUCell.CellId to associate it with the NR cell  
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,  
PhysicalCellId=1, FrequencyBand=N77, DfNarfcn=636666, UfBandwidth=CELL_BW_100M,  
DfBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSU, SlotStructure=SS2,  
LampSiteCellFlag=YES, LogicalRootSequenceIndex=80, TrackingAreaId=0,  
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=7881, SsbSubcarrierSpacing=30KHZ;  
//Adding an NR DU cell TRP, setting NRDUCellTrp.NrDuCellId to specify the NR DU cell served by this TRP,  
and setting NRDUCellTrp.BasebandEqmId to bind baseband equipment to the NR DU cell TRP (If no  
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell  
TRP.)  
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=4T4R, BasebandEqmId=0,  
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=240, CpriCompression=3DOT2_COMPRESSION,  
FrAndDuplexMode=FR1_TDD;  
//Adding an NR DU cell coverage area, setting NRDUCellCoverage.NrDuCellTrpId to specify the NR DU cell  
TRP corresponding to this coverage area, and setting NRDUCellCoverage.SectorEqmId to bind sector  
equipment to the NR DU cell TRP  
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;  
//Activating the cell  
ACT NRCELL: NrCellId=0;
```

Establishing a 2T2R TDD cell (using a 100 MHz cell as an example)

```
//Adding a sector and sector equipment, with the SECTOR.ANTNUM parameter set to 2, and configuring  
two antenna channels of an RRU for this sector  
ADD SECTOR: SECTORID=0, SECNAME="0", USERLABEL="0", ANTNUM=2, ANT1CN=0, ANT1SRN=60,  
ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, CREATESECTOREQM=TRUE,  
SECTOREQMID=0;  
//Adding baseband equipment  
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=0;  
  
//(Optional) Adding operator and tracking area information (if not added previously)  
//Adding an operator "XXXX"  
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",  
NrNetworkingOption=SA;  
//Adding a tracking area  
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;  
  
//Adding a 100 MHz TDD cell  
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N78, DuplexMode=CELL_TDD;
```

```
//Adding an NR DU cell, and setting NRDUCell.CellId to associate it with the NR cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N78, DINarfcn=634000, UIBandwidth=CELL_BW_100M,
DIBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7853, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP, setting NRDUCellTrp.NrDuCellId to specify the NR DU cell served by this TRP,
and setting NRDUCellTrp.BasebandEqmId to bind baseband equipment to the NR DU cell TRP (If no
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell
TRP)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=2T2R,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=320, BasebandEqmId=0,
CpriCompression=3DOT2_COMPRESSION, FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, setting NRDUCellCoverage.NrDuCellTrpId to specify the NR DU cell
TRP corresponding to this coverage area, and setting NRDUCellCoverage.SectorEqmId to bind sector
equipment to the NR DU cell TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;
ACT NRCELL: NrCellId=0;
```

Establishing a 1T1R TDD cell (using a 100 MHz cell as an example)

```
//Adding a sector and sector equipment, with the SECTOR.ANTNUM parameter set to 1, and configuring
one antenna channel of an RRU for this sector
ADD SECTOR: SECTORID=0, SECNAME="0", USERLABEL="0", ANTNUM=1, ANT1CN=0, ANT1SRN=60,
ANT1SN=0, ANT1N=R0A, CREATESECTOREQM=TRUE, SECTOREQMID=0;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding a 100 MHz TDD cell
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N78, DuplexMode=CELL_TDD;
//Adding an NR DU cell, and setting NRDUCell.CellId to associate it with the NR cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N78, DINarfcn=634000, UIBandwidth=CELL_BW_100M,
DIBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7853, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP, setting NRDUCellTrp.NrDuCellId to specify the NR DU cell served by this TRP,
and setting NRDUCellTrp.BasebandEqmId to bind baseband equipment to the NR DU cell TRP (If no
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell
TRP)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=1T1R,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=320, BasebandEqmId=0,
CpriCompression=3DOT2_COMPRESSION, FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, setting NRDUCellCoverage.NrDuCellTrpId to specify the NR DU cell
TRP corresponding to this coverage area, and setting NRDUCellCoverage.SectorEqmId to bind sector
equipment to the NR DU cell TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;
ACT NRCELL: NrCellId=0;
```

3.4.1.2.2 Establishing Cells in a Multi-Carrier Sector

NOTICE

- Signals over other carriers will be interrupted intermittently the moment a carrier is added to or removed from a multi-carrier RRU. Considering this, you are not advised to add a carrier, remove a carrier, or reset a cell (through removing a carrier and then adding a carrier) during peak hours.
- Other carriers will be reestablished the moment a carrier is added to or removed from a multi-carrier high-frequency module. UEs served by these carriers will experience service drops. Considering this, you are not advised to perform operations such as add or remove a carrier during peak hours.
- Before establishing cells in a multi-carrier sector, ensure that the multi-carrier capabilities of RRUs and CPRI ports and the cell specifications of the BBP meet the requirements described in [3.3.3 Hardware](#).

Establishing TDD cells in a two-carrier sector in low frequency bands

```
//Adding a sector and sector equipment
ADD SECTOR: SECTORID=0, SECNAME="0", LOCATIONNAME="0", USERLABEL="0", ANTNUM=4,
ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B,
ANT3CN=0, ANT3SRN=60, ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D,
CREATESECTOREQM=TRUE, SECTOREQMID=0;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding cell 1
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N77, DuplexMode=CELL_TDD;
//Adding an NR DU cell for cell 1, with NRDUCell.CellId specifying the associated NR cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", CellId=1, PhysicalCellId=0,
FrequencyBand=N77, DfNarfcn=623000, UlBandwidth=CELL_BW_40M, DlBandwidth=CELL_BW_40M,
DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7739, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP for cell 1, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by
this TRP and NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP (If no
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell
TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=4T4R, BasebandEqmId=0,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=320, CpriCompression=3DOT2_COMPRESSION,
FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell
TRP corresponding to this coverage area and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;
//Adding cell 2
ADD NRCELL: NrCellId=1, CellName="CELL2", CellId=2, FrequencyBand=N77, DuplexMode=CELL_TDD;
//Adding an NR DU cell for cell 2, with NRDUCell.CellId specifying the associated NR cell and
NRDUCell.DfNarfcn set to a different value than that for cell 1 to ensure that the frequency ranges of the
two cells do not overlap
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="DUCELL2", CellId=2, PhysicalCellId=0,
FrequencyBand=N77, DfNarfcn=625666, UlBandwidth=CELL_BW_40M, DlBandwidth=CELL_BW_40M,
DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=92, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7766, SsbSubcarrierSpacing=30KHZ;
```

```
//Adding an NR DU cell TRP for cell 2, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by
this TRP, NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP, and
NRDUCellCoverage.SectorEqmId set to the same value as that for cell 1
ADD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=1, TxRxMode=4T4R, BasebandEqmId=0,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=0, CpriCompression=3DOT2_COMPRESSION,
FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell
TRP corresponding to this coverage area and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=0, SectorEqmId=0;
//Activating the cells
ACT NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=1;
```

Establishing TDD cells in a two-carrier sector in high frequency bands

```
//Adding a sector
ADD SECTOR: SECTORID=1, ANTNUM=0, CREATESECTOREQM=FALSE;
//Adding sector equipment, with SectorEqm.SectorId specifying the sector associated with the sector
equipment
ADD SECTOREQM: SECTOREQMID=1, SECTORID=1, ANTCFGMODE=BEAM, RRUCN=0, RRUSRN=63,
RRUSN=0, BEAMSHAPE=SEC_120DEG, BEAMLAYERSPLIT=None, BEAMAZIMUTHOFFSET=None;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=2, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=2;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="262", Mnc="03",
NrNetworkingOption=NSA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=2;

//Adding cells 1 and 2
ADD NRCELL: NrCellId=101, CellName="101", CellId=101, FrequencyBand=N257, DuplexMode=CELL_TDD;
ADD NRCELL: NrCellId=102, CellName="102", CellId=102, FrequencyBand=N257, DuplexMode=CELL_TDD;
//Adding NR DU cells for cells 1 and 2, and ensuring that the frequency ranges of the cells do not overlap
(with NRDUCell.DlNarfcn set to different values). Except for the parameters of basic cell attributes, the
settings of other parameters must be identical for the cells.
ADD NRDUCELL: NrDuCellId=101, NrDuCellName="101", DuplexMode=CELL_TDD, CellId=101,
PhysicalCellId=101, FrequencyBand=N257, DlNarfcn=2079957, UlBandwidth=CELL_BW_200M,
DlBandwidth=CELL_BW_200M, SubcarrierSpacing=120KHZ, SlotAssignment=4_1_DDSU,
SlotStructure=SS202, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=22476,
LogicalRootSequenceIndex=80, SsbSubcarrierSpacing=120KHZ;
ADD NRDUCELL: NrDuCellId=102, NrDuCellName="102", DuplexMode=CELL_TDD, CellId=102,
PhysicalCellId=101, FrequencyBand=N257, DlNarfcn=2083289, UlBandwidth=CELL_BW_200M,
DlBandwidth=CELL_BW_200M, SubcarrierSpacing=120KHZ, SlotAssignment=4_1_DDSU,
SlotStructure=SS202, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=22487,
LogicalRootSequenceIndex=80, SsbSubcarrierSpacing=120KHZ;
//Adding NR DU cell TRPs for cells 1 and 2, with NRDUCellTrp.NrDuCellId specifying the NR DU cells served
by these TRPs and NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to these TRPs
(If no baseband equipment is bound, the system automatically selects baseband equipment for the NR DU
cell TRPs.)
ADD NRDUCELLTRP: NrDuCellTrpId=101, NrDuCellId=101, TxRxMode=4T4R, BasebandEqmId=2,
PowerConfigMode=EIRP, MaxEirp=640, CpriCompression=2DOT5_COMPRESSION,
FrAndDuplexMode=FR2_TDD;
ADD NRDUCELLTRP: NrDuCellTrpId=102, NrDuCellId=102, TxRxMode=4T4R, BasebandEqmId=2,
PowerConfigMode=EIRP, MaxEirp=640, CpriCompression=2DOT5_COMPRESSION,
FrAndDuplexMode=FR2_TDD;
//Adding NR DU cell coverage areas for cells 1 and 2, with NRDUCellCoverage.NrDuCellTrpId specifying the
NR DU cell TRPs corresponding to these coverage areas and NRDUCellCoverage.SectorEqmId specifying the
sector equipment bound to these TRPs
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=101, NrDuCellCoverageId=101, SectorEqmId=1;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=102, NrDuCellCoverageId=102, SectorEqmId=1;
//Activating the cells
ACT NRCELL: NrCellId=101;
ACT NRCELL: NrCellId=102;
```

Establishing TDD cells in a four-carrier sector in high frequency bands (using 100 MHz per carrier as an example)

```
//Adding a sector
ADD SECTOR: SECTORID=1, ANTNUM=0, CREATESECTOREQM=FALSE;
//Adding sector equipment, with SectorEqm.SectorId specifying the sector associated with the sector
equipment
ADD SECTOREQM: SECTOREQMID=1, SECTORID=1, ANTCFGMODE=BEAM, RRUCN=0, RRUSRN=63,
RRUSN=0, BEAMSHAPE=SEC_120DEG, BEAMLAYERSPLIT=None, BEAMAZIMUTHOFFSET=None;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=2;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="262", Mnc="03",
NrNetworkingOption=NSA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=2;

//Adding cells 1 to 4
ADD NRCELL: NrCellId=11, CellName="11", CellId=11, FrequencyBand=N257, DuplexMode=CELL_TDD;
ADD NRCELL: NrCellId=12, CellName="12", CellId=12, FrequencyBand=N257, DuplexMode=CELL_TDD;
ADD NRCELL: NrCellId=13, CellName="13", CellId=13, FrequencyBand=N257, DuplexMode=CELL_TDD;
ADD NRCELL: NrCellId=14, CellName="14", CellId=14, FrequencyBand=N257, DuplexMode=CELL_TDD;
//Adding NR DU cells for cells 1 to 4, and ensuring that the frequency ranges of the cells do not overlap
(with NRDUCell.DINarfcn set to different values). Except for the parameters of basic cell attributes, the
settings of other parameters must be identical for the cells.
ADD NRDUCELL: NrDuCellId=11, NrDuCellName="11", DuplexMode=CELL_TDD, CellId=11,
PhysicalCellId=11, FrequencyBand=N257, DINarfcn=2074959, ULBandwidth=CELL_BW_100M,
DLBandwidth=CELL_BW_100M, CellRadius=800, SubcarrierSpacing=120KHZ, SlotAssignment=4_1_DDSSU,
SlotStructure=SS202, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=22458,
LogicalRootSequenceIndex=80, SupplementaryCellInd=NORMAL_CELL, SsbSubcarrierSpacing=120KHZ;
ADD NRDUCELL: NrDuCellId=12, NrDuCellName="12", DuplexMode=CELL_TDD, CellId=12,
PhysicalCellId=11, FrequencyBand=N257, DINarfcn=2076625, ULBandwidth=CELL_BW_100M,
DLBandwidth=CELL_BW_100M, CellRadius=800, SubcarrierSpacing=120KHZ, SlotAssignment=4_1_DDSSU,
SlotStructure=SS202, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=22464,
LogicalRootSequenceIndex=80, SupplementaryCellInd=NORMAL_CELL, SsbSubcarrierSpacing=120KHZ;
ADD NRDUCELL: NrDuCellId=13, NrDuCellName="13", DuplexMode=CELL_TDD, CellId=13,
PhysicalCellId=11, FrequencyBand=N257, DINarfcn=2078291, ULBandwidth=CELL_BW_100M,
DLBandwidth=CELL_BW_100M, CellRadius=800, SubcarrierSpacing=120KHZ, SlotAssignment=4_1_DDSSU,
SlotStructure=SS202, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=22470,
LogicalRootSequenceIndex=80, SupplementaryCellInd=NORMAL_CELL, SsbSubcarrierSpacing=120KHZ;
ADD NRDUCELL: NrDuCellId=14, NrDuCellName="14", DuplexMode=CELL_TDD, CellId=14,
PhysicalCellId=11, FrequencyBand=N257, DINarfcn=2079957, ULBandwidth=CELL_BW_100M,
DLBandwidth=CELL_BW_100M, CellRadius=800, SubcarrierSpacing=120KHZ, SlotAssignment=4_1_DDSSU,
SlotStructure=SS202, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=22476,
LogicalRootSequenceIndex=80, SupplementaryCellInd=NORMAL_CELL, SsbSubcarrierSpacing=120KHZ;
//Adding NR DU cell TRPs for cells 1 to 4, with NRDUCellTrp.NrDuCellId specifying the NR DU cells served
by these TRPs and NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to these TRPs
(If no baseband equipment is bound, the system automatically selects baseband equipment for the NR DU
cell TRPs.)
ADD NRDUCELLTRP: NrDuCellTrpId=11, NrDuCellId=11, TxRxMode=4T4R, BasebandEqmId=0,
PowerConfigMode=EIRP, MaxEirp=610, CpriCompression=2DOT5_COMPRESSION,
FrAndDuplexMode=FR2_TDD;
ADD NRDUCELLTRP: NrDuCellTrpId=12, NrDuCellId=12, TxRxMode=4T4R, BasebandEqmId=0,
PowerConfigMode=EIRP, MaxEirp=610, CpriCompression=2DOT5_COMPRESSION,
FrAndDuplexMode=FR2_TDD;
ADD NRDUCELLTRP: NrDuCellTrpId=13, NrDuCellId=13, TxRxMode=4T4R, BasebandEqmId=0,
PowerConfigMode=EIRP, MaxEirp=610, CpriCompression=2DOT5_COMPRESSION,
FrAndDuplexMode=FR2_TDD;
ADD NRDUCELLTRP: NrDuCellTrpId=14, NrDuCellId=14, TxRxMode=4T4R, BasebandEqmId=0,
PowerConfigMode=EIRP, MaxEirp=610, CpriCompression=2DOT5_COMPRESSION,
FrAndDuplexMode=FR2_TDD;
//Adding NR DU cell coverage areas for cells 1 to 4, with NRDUCellCoverage.NrDuCellTrpId specifying the
NR DU cell TRPs corresponding to these coverage areas and NRDUCellCoverage.SectorEqmId specifying the
sector equipment bound to these TRPs
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=11, NrDuCellCoverageId=11, SectorEqmId=1;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=12, NrDuCellCoverageId=12, SectorEqmId=1;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=13, NrDuCellCoverageId=13, SectorEqmId=1;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=14, NrDuCellCoverageId=14, SectorEqmId=1;
//Activating the cells
ACT NRCELL: NrCellId=11;
```


ACT NRCELL: NrCellId=12;
ACT NRCELL: NrCellId=13;
ACT NRCELL: NrCellId=14;

3.4.1.2.3 Establishing an SUL Cell

NOTE

An SUL cell corresponds to one NR cell and two NR DU cells.

```
//For the TDD cell
//Adding a sector
ADD SECTOR: SECTORID=0, SECNAME="0", LOCATIONNAME="0", USERLABEL="0", ANTNUM=0,
CREATESECTOREQM=FALSE;
//Adding sector equipment, with SectorEqm.SectorId specifying the sector associated with the sector
equipment
ADD SECTOREQM: SECTOREQMID=0, SECTORID=0, ANTFCGMode=BEAM, RRUCN=0, RRUSRN=60,
RRUSN=0, BEAMSHAPE=SEC_120DEG, BEAMLAYERSPLIT=None, BEAMAZIMUTHOFFSET=None;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=UDDL, UMTSDEMMode=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding a 100 MHz TDD cell
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N78, DuplexMode=CELL_TDD,
UserLabel="CELL1";
//Adding an NR DU cell, with NRDUCell.CellId specifying the associated NR cell
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="CELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N78, DfNarfcn=634000, UfBandwidth=CELL_BW_100M,
DfBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7853, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by this TRP and
NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP (If no baseband
equipment is bound, the system automatically selects baseband equipment for the NR DU cell TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=1, TxRxMode=64T64R, BasebandEqmId=0,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=300, CpriCompression=3DOT2_COMPRESSION,
FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell
TRP corresponding to this coverage area and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;

//For the SUL cell
//Adding a sector
ADD SECTOR: SECTORID=1, SECNAME="1", USERLABEL="1", ANTNUM=4, ANT1CN=0, ANT1SRN=61,
ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=61, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=61,
ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=61, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=FALSE;
//Adding sector equipment, with SectorEqm.SectorId specifying the sector associated with the sector
equipment
ADD SECTOREQM: SECTOREQMID=1, SECTORID=1, ANTFCGMode=ANTENNAPORT, ANTNUM=4,
ANT1CN=0, ANT1SRN=61, ANT1SN=0, ANT1N=R0A, ANTTYPE1=RX_MODE, ANT2CN=0, ANT2SRN=61,
ANT2SN=0, ANT2N=R0B, ANTTYPE2=RX_MODE, ANT3CN=0, ANT3SRN=61, ANT3SN=0, ANT3N=R0C,
ANTTYPE3=RX_MODE, ANT4CN=0, ANT4SRN=61, ANT4SN=0, ANT4N=R0D, ANTTYPE4=RX_MODE;
//Adding an SUL DU cell
ADD NRDUCELL: NrDuCellId=2, NrDuCellName="SUL", DuplexMode=CELL_SUL, FrequencyBand=N80,
UfNarfcn=350000, UfBandwidth=CELL_BW_20M, LogicalRootSequenceIndex=92,
DeployPosition=DEPLOY_INTEGRATED;
//Adding an NR DU cell TRP, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by this TRP and
NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP (If no baseband
equipment is bound, the system automatically selects baseband equipment for the NR DU cell TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=2, TxRxMode=4R, BasebandEqmId=0,
CpriCompression=NO_COMPRESSION, FrAndDuplexMode=SUL;
//Adding an NR DU cell coverage area, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell
```

```
TRP corresponding to this coverage area and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=0, SectorEqmId=1;
//Adding an SUL relationship for the cell
ADD NRDUCELLSULRELATION: NrDuCellId=1, SulIndex=0, SulType=FIX_SUL, SulNrDuCellId=2;
//Turning on the UL and DL Decoupling switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, ULDecouplingSwitch=ON;
//Activating the cell
ACT NRCELL: NrCellId=0;
//((Optional, required only when SUL is enabled for an existing NR TDD cell) Redeploying the NR DU cell.
This command involves the reestablishment of cells served by the BBP. You are advised to run this
command when traffic is light (for example, in the early morning).
STR NRDUCELLREDEPLOY: DeployType=CONFLICT;
```

3.4.1.2.4 Establishing a Combined-RRU Cell

Establishing a 4R SUL cell (using a 20 MHz cell served by two combined 2R RRUs as an example)

NOTE

This section uses the process for establishing a combined-RRU SUL cell as an example. For details about the MML command examples for establishing a TDD cell, see [3.4.1.2.1 Establishing a TDD Cell](#).

```
//Adding a sector
ADD SECTOR: SECTORID=1, SECNAME="1", USERLABEL="1", ANTNUM=4, ANT1CN=0, ANT1SRN=61,
ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=61, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=62,
ANT3SN=0, ANT3N=R0A, ANT4CN=0, ANT4SRN=62, ANT4SN=0, ANT4N=R0B, CREATESECTOREQM=FALSE;
//Adding sector equipment, with SectorEqm.SectorId specifying the sector associated with the sector
equipment
ADD SECTOREQM: SECTOREQMID=1, SECTORID=1, ANTFCGMODE=ANTENNAPORT, ANTNUM=4,
ANT1CN=0, ANT1SRN=61, ANT1SN=0, ANT1N=R0A, ANTTYPE1=RX_MODE, ANT2CN=0, ANT2SRN=61,
ANT2SN=0, ANT2N=R0B, ANTTYPE2=RX_MODE, ANT3CN=0, ANT3SRN=62, ANT3SN=0, ANT3N=R0A,
ANTTYPE3=RX_MODE, ANT4CN=0, ANT4SRN=62, ANT4SN=0, ANT4N=R0B, ANTTYPE4=RX_MODE;
//Adding an SUL DU cell
ADD NRDUCELL: NrDuCellId=8, NrDuCellName="SUL", DuplexMode=CELL_SUL, FrequencyBand=N80,
UINarfcn=350000, UIBandwidth=CELL_BW_20M, LogicalRootSequenceIndex=92,
DeployPosition=DEPLOY_INTEGRATED;
//Adding an NR DU cell TRP, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by this TRP and
NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP (If no baseband
equipment is bound, the system automatically selects baseband equipment for the NR DU cell TRP)
ADD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=2, TxRxMode=4R, BasebandEqmId=0,
CpriCompression=NO_COMPRESSION, FrAndDuplexMode=SUL;
//Adding an NR DU cell coverage area, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell
TRP corresponding to this coverage area and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=0, SectorEqmId=1;
//Adding an SUL relationship for the cell
ADD NRDUCELLSULRELATION: NrDuCellId=1, SulIndex=0, SulType=FIX_SUL, SulNrDuCellId=2;
//Turning on the UL and DL Decoupling switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, ULDecouplingSwitch=ON;
//Activating the cell
ACT NRCELL: NrCellId=0;
//((Optional, required only when SUL is enabled for an existing NR TDD cell) Redeploying the NR DU cell.
This command involves the reestablishment of cells served by the BBP. You are advised to run this
command when traffic is light (for example, in the early morning).
STR NRDUCELLREDEPLOY: DeployType=CONFLICT;
```

3.4.1.2.5 Establishing Split-Sector Cells

Establishing two split-sector 2T2R cells

```
//Configuring two sectors with the SECTOR.ANTNUM parameter set to 2, and then allocating any two
antenna channels of the RRU to one sector and the other two antenna channels to the other sector (For
example, after two 2T2R sectors numbered 0 and 1 and related sector equipment are configured, sector 0
uses antenna channels A and B, and sector 1 uses antenna channels C and D.)
ADD SECTOR: SECTORID=0, SECNAME="0", LOCATIONNAME="0", USERLABEL="0", ANTNUM=2,
```

```
ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B,
CREATESECTOREQM=TRUE, SECTOREQMID=0;
ADD SECTOR: SECTORID=1, SECNAME="0", LOCATIONNAME="0", USERLABEL="0", ANTNUM=2,
ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0C, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0D,
CREATESECTOREQM=TRUE, SECTOREQMID=1;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMODE=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX"
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding cell 1
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N77, DuplexMode=CELL_TDD;
//Adding an NR DU cell for cell 1, with NRDUCell.CellId specifying the associated NR cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", CellId=1, PhysicalCellId=0,
FrequencyBand=N77, DlnArfcn=623000, UlBandwidth=CELL_BW_40M, DlBandwidth=CELL_BW_40M,
DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7739, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP for cell 1, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by
this TRP and NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP (If no
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell
TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=2T2R, BasebandEqmId=0,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=320, CpriCompression=3DOT2_COMPRESSION,
FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell
TRP corresponding to this coverage area and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;
//Adding cell 2
ADD NRCELL: NrCellId=1, CellName="CELL2", CellId=2, FrequencyBand=N77, DuplexMode=CELL_TDD;
//Adding an NR DU cell for cell 2, with NRDUCell.CellId specifying the associated NR cell
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="DUCELL2", CellId=2, PhysicalCellId=1,
FrequencyBand=N77, DlnArfcn=623000, UlBandwidth=CELL_BW_40M, DlBandwidth=CELL_BW_40M,
DuplexMode=CELL_TDD, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDSSU, SlotStructure=SS2,
LogicalRootSequenceIndex=80, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7739, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP for cell 2, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by
this TRP and NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP (If no
baseband equipment is bound, the system automatically selects baseband equipment for the NR DU cell
TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=1, TxRxMode=2T2R, BasebandEqmId=0,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=320, CpriCompression=3DOT2_COMPRESSION,
FrAndDuplexMode=FR1_TDD;
//Adding an NR DU cell coverage area, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell
TRP corresponding to this coverage area and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=0, SectorEqmId=1;
//Activating the cells
ACT NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=1;
```

3.4.1.2.6 Establishing an RF Combination Cell

Establishing a LampSite 4T4R TDD RF combination cell (using a 100 MHz cell as an example)

```
//Adding sectors and pieces of sector equipment, with the SECTOR.ANTNUM parameter set to 4 (An RF
combination cell requires multiple sectors and pieces of sector equipment.)
ADD SECTOR: SECTORID=0, SECNAME="0", USERLABEL="0", ANTNUM=4, ANT1CN=0, ANT1SRN=60,
ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=60,
ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,
SECTOREQMID=0;
ADD SECTOR: SECTORID=1, SECNAME="1", USERLABEL="1", ANTNUM=4, ANT1CN=0, ANT1SRN=61,
```

```

ANT1SN=0, ANT1N=R0A, ANT2CN=0, ANT2SRN=61, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=61,
ANT3SN=0, ANT3N=R0C, ANT4CN=0, ANT4SRN=61, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,
SECTOREQMID=1;
//Adding baseband equipment
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=0;

//(Optional) Adding operator and tracking area information (if not added previously)
//Adding an operator "XXXX", with the NR networking option set to SA or NSA
ADD GNBOPERATOR: OperatorId=0, OperatorName="XXXX", Mcc="460", Mnc="00",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA;
//Adding a tracking area
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;

//Adding an NR cell
ADD NRCELL: NrCellId=0, CellName="CELL1", CellId=1, FrequencyBand=N77, DuplexMode=CELL_TDD;
//Adding an NR DU cell, with NRDUCell.CellId specifying the associated NR cell
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="DUCELL1", DuplexMode=CELL_TDD, CellId=1,
PhysicalCellId=1, FrequencyBand=N77, DLNarfcn=636666, ULBandwidth=CELL_BW_100M,
DLBandwidth=CELL_BW_100M, SubcarrierSpacing=30KHZ, SlotAssignment=4_1_DDDSU, SlotStructure=SS2,
LampSiteCellFlag=YES, LogicalRootSequenceIndex=80, TrackingAreaId=0,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=7881, SsbSubcarrierSpacing=30KHZ;
//Adding an NR DU cell TRP, with NRDUCellTrp.NrDuCellId specifying the NR DU cell served by this TRP and
NRDUCellTrp.BasebandEqmId specifying the baseband equipment bound to this TRP (If no baseband
equipment is bound, the system automatically selects baseband equipment for the NR DU cell TRP.)
ADD NRDUCELLTRP: NrDuCellTrpId=0, NrDuCellId=0, TxRxMode=4T4R, BasebandEqmId=0,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=240, CpriCompression=3DOT2_COMPRESSION,
FrAndDuplexMode=FR1_TDD;
//Adding NR DU cell coverage areas, with NRDUCellCoverage.NrDuCellTrpId specifying the NR DU cell TRP
corresponding to these coverage areas and NRDUCellCoverage.SectorEqmId specifying the sector
equipment bound to this TRP (An RF combination cell requires multiple NR DU cell coverage areas, with
each bound to one piece of sector equipment.)
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=0, SectorEqmId=0;
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=1, SectorEqmId=1;
//Activating the cell
ACT NRCELL: NrCellId=0;

```

3.4.1.2.7 Other Configurations Related to Cell Management

The configurations in the following examples are for reference only. On live networks, configure the parameters based on the network plan.

```

//Performing high-priority NR DU cell blocking
BLK NRDUCELL: NrDuCellId=0, CellAdminState=CELL_HIGH_BLOCK;
//Setting the ServiceTypeTag parameter to TOB
MOD NRCELLLOAMPARAM: NrCellId=0, ServiceTypeTag=TOB;
//Setting the NrDuCellNetworkingMode parameter to NORMAL_CELL
MOD NRDUCELL: NrDuCellId=0, NrDuCellNetworkingMode=NORMAL_CELL;

```

3.4.1.2.8 Deactivating a Cell

```

//Deactivating a cell
DEA NRCELL: NrCellId=0;

```

3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.2 Activation Verification

Step 1 Run the **DSP NRCELL** command to query the cell status.

If the value of **Cell Available State** in the command output is **Available**, the cell has been successfully set up.

Step 2 Verify cell functions.

Run the **DSP NRDUCELLUENUMBER** command to query the cell status.

If the value of **High-Reliability RRC_CONNECTED UE Number** in the command output is not **0**, UEs successfully access the cell and the cell is working properly.

----End

3.4.3 Network Monitoring

None

4 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

5 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.401: "NG-RAN; Architecture description"
- 3GPP TR 38.913: "Study on scenarios and requirements for next generation access technologies"
- 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)"
- *RF Unit and Topology Management*
- *CPRI Compression*
- *UL and DL Decoupling*
- *Hyper Cell*
- *Cell Combination*
- *Standards Compliance*
- *License Control Item Lists*
- *Scalable Bandwidth*
- *IPv4 Transmission*
- *Synchronization*
- *IPv6 Transmission*
- *CPRI MUX*
- *Beam Management*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *gNodeB Initial Configuration Guide (MML-based) in 3900 & 5900 Series Base Station Product Documentation*
- *ALD Management in 3900 & 5900 Series Base Station Product Documentation*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*